

Seeing Seizures: A Large-Scale Empirical Study of Video and EEG Deep Learning for Automated Rodent Seizure Detection and Severity Grading

Anonymous WACV Applications Track submission

Paper ID *****

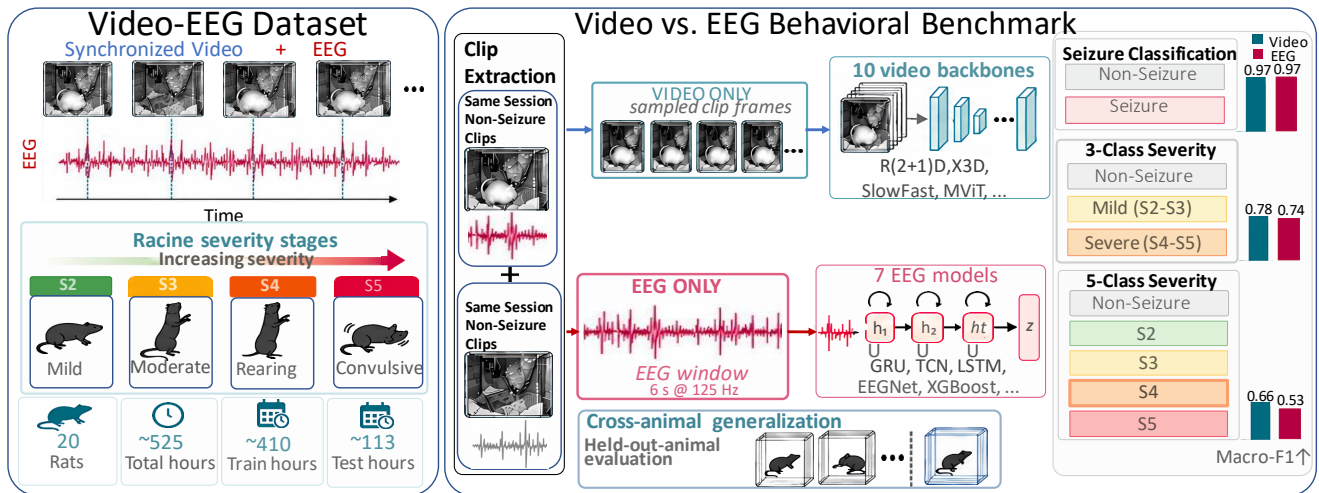


Figure 1. Overview of the study. *Left*: the cohort—synchronized video and EEG from 20 rats (corpus scale in App. C.1, Table 12), each clip graded on the Racine scale from Stage 2 (mild) to Stage 5 (convulsive). *Right*: the benchmark. Expert-annotated seizure clips and same-session non-seizure clips feed two independent, unfused lanes—ten video backbones and seven EEG models—compared head-to-head at 2-, 3- and 5-class granularity and under held-out-animal evaluation. Bars report macro-F1: the two modalities are level at detection and separate as the task becomes explicitly behavioral.

Abstract

001 Video and EEG are widely used to study seizures in animals.
002 However, manual scoring remains a huge bottleneck. Mak-
003 ing this automatic is a big challenge that few have solved
004 for animal models. This study tackles the problem using
005 data from twenty rats, analyzing about 24000 paired clips
006 from 601 sessions. Experts graded the seizures from a mild
007 Stage 2 to a severe Stage 5. Ten video models and seven EEG
008 models were tested under strict rules to see how well they
009 could spot seizures and grade their exact severity. For basic
010 detection, both tools did well, while EEG was slightly better,
011 scoring 0.974 compared to video’s 0.968. Both tools also
012 worked well on unseen rats. The gap grew when grading
013 the exact seizure levels. In the five-level test, video scored
014 0.657, easily beating EEG’s 0.560. The two methods also
015 made different types of mistakes. Video usually just con-
016 fused stages that were next to each other. EEG was more

likely to miss seizures entirely or mix up very different stages.
The size of the video model did not matter much. A tiny
model did almost as well as a huge one. Instead, tracking
motion over time was the real key. An additional test on
the outside RodEpi dataset confirmed these results, with the
model score reaching 0.979. Overall, we demonstrate that
cameras offer a great, pain-free way to track animal seizures
and work very well alongside typical EEG tools.

1. Introduction

Epilepsy affects roughly 1% of the world’s population [33].
Almost everything we know about its mechanisms, and ev-
ery candidate anti-seizure therapy, is first established in an-
imal models, predominantly chemically or electrically in-
duced rodents [36]. These studies rest on *synchronized*
video-EEG: an overhead camera records behavior while im-
planted electrodes record cortical activity. The two modal-



Figure 2. A representative overhead frame of the acquisition setup: each rat is recorded in its home cage by a fixed overhead camera while implanted electrodes stream synchronized EEG, and the cage rests on a per-animal platform. Camera geometry differs per cage, so clips are per-cage cropped before subjects are pooled. De-identified (cage-side identifier redacted).

ities are complementary. Electrographic seizures can occur with little visible movement, and movement artifacts routinely corrupt the EEG, so an experimenter confirms that an event is a seizure, and grades its behavioral severity on the Racine scale, only by reading both channels together [43, 58]. This complementarity makes video-EEG the gold standard of preclinical epilepsy research [36].

Manual scoring makes this rigor expensive. A single study can generate thousands of hours of paired video and EEG, and a trained annotator reviews all of it. Automating the behavioral half of this workflow is a *video-understanding* problem: modern action-recognition backbones [10, 17, 51] and animal pose estimators [38, 41] are mature and transferable. Yet the epilepsy literature has been slow to adopt them, especially in animal studies. Video-only rodent detection has only just acquired its first public benchmark [42], and computational video-EEG fusion in animals is confined to two isolated *binary* detectors [35, 40], neither grading severity. No one has yet measured *how far standard video deep learning actually gets* on this problem against a strong EEG reference on the same data.

This paper provides that account. The contribution is empirical: on a real preclinical cohort, under a single controlled protocol, we report where a standard video backbone matches EEG, where it surpasses it, where it breaks, and how the choice of architecture, temporal sampling, data budget and evaluation protocol shape the answer. A comprehensive literature survey of the field (video-only and multimodal video-EEG detection across human and animal subjects) is provided in the supplementary material (App. A); the main paper is the empirical study.

We organize that study around four questions:

RQ1 *What model and input capacity is sufficient?* (Sec. 5.1)

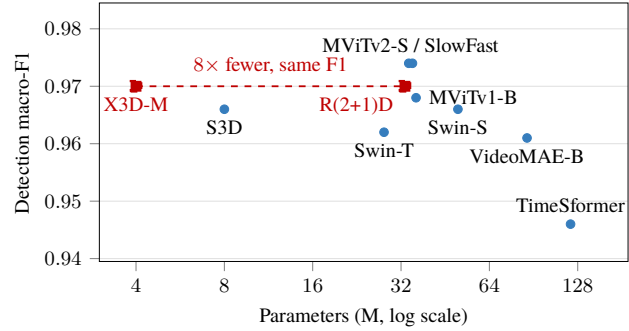


Figure 3. Detection accuracy vs. model size across ten video backbones (accuracies from Table 2, parameter counts from Table 9). Every architecture lands in a narrow 0.946–0.974 macro-F1 band, and the 4M-parameter X3D-M ties the 33M R(2+1)D-18 at 0.970, an 8× saving at no cost in accuracy.

RQ2 *Do the two modalities fail in the same way, or are their errors complementary?* (Sec. 5.2)

RQ3 *Once a seizure is identified, does ordinal modeling improve Stage 2–5 grading and reduce large-stage errors on unseen subjects?* (Sec. 3.2)

RQ4 *Does inverse-frequency weighting keep the long tail from dominating the result?* (Sec. 5.3)

All four rest on the head-to-head comparison itself: how far a standard video backbone gets against an electrode-based reference, and whether that ordering survives held-out animals and an external benchmark (Sec. 5.2). Our contributions are:

- A **pooled twenty-subject synchronized video-EEG cohort** (Sec. 4): ~24k expert-annotated, per-cage-cropped clips over 601 sessions with per-clip Racine severity labels and severely long-tailed burden. To our knowledge this is the largest paired rodent video-EEG corpus studied for learned detection. Figure 2 shows the acquisition setup: one fixed overhead camera per home cage, so geometry is constant within an animal and differs across animals, which is why we crop per cage. We release a fully de-identified description.
- A **head-to-head video-vs.-EEG comparison** across three task granularities (Sec. 5) under a matched, leakage-controlled protocol. Video matches a strong EEG reference on detection (0.993 vs. 0.992 AUROC) and degrades far more gracefully on severity grading, with Grad-CAM evidence that the model attends to behaviorally meaningful regions.
- A **controlled sweep of ten video backbones and seven EEG models**, plus temporal-sampling, resolution, data-efficiency and pretraining ablations. Detection is backbone-agnostic on this cohort (Fig. 3): accuracy is flat in model size, and a 4M-parameter X3D matches a 33M R(2+1)D. Temporal context is what matters: a single-

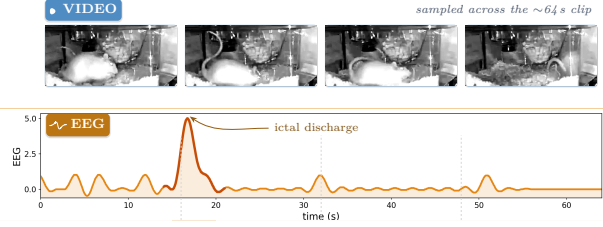


Figure 4. Synchronized video-EEG for one Stage-3 seizure clip (subject R7). *Top*: four frames sampled across the clip (contrast-enhanced), aligned to the shared time axis. *Bottom*: the simultaneous single-channel EEG (~64s), the shaded window marking the ictal discharge. Video carries the convulsive motion, EEG the electrographic discharge; this pairing is what our study rests on. De-identified.

frame control collapses at every granularity (0.569 macro-F1 at detection against 0.970 at 16 frames), which shows the task is read from motion. Under **subject-disjoint** cross-validation the two modalities stay level at detection, while video’s advantage at 3- and 5-class grading holds.

- **External validation** on the public RodEpile benchmark [42] (0.979 subject-wise F1), together with a documented encoding artifact in its raw release that must be corrected for an honest score.

2. Related Work

Efforts to automate seizure scoring divide by the signal they read. We position our study against three such lines, each treated at length in the supplementary survey (App. A). That survey is organized by modality, covering EEG-only (App. A.2), video-only (App. A.3) and fused video-EEG (App. A.4) work, and by subject (human vs. animal). Its gap analysis (App. A.1, Table 7) locates our contribution.

EEG-only detection is the incumbent and sets the accuracy bar. Rodent systems have moved from hand-crafted spectral features with classical classifiers [6] to end-to-end deep networks: spectrogram CNNs over thousands of hours [27] and recurrent GRU/LSTM models that outperform Random Forest and XGBoost on chemically diverse cohorts [18]. All operate on EEG alone and discard the synchronized video recorded alongside it. We adopt this family as a deliberately strong seven-model EEG reference for our cross-modal comparison: a GRU, an LSTM, EEGNet, an EEG Conformer, a TCN, and the two classical baselines.

Video-only seizure detection reframes the task as action recognition, inheriting mature video-understanding backbones (two-stream, 3D and (2+1)D convolutions, and video transformers [7, 10, 17, 48, 50, 51]; reviewed in App. A.3.1) and animal pose estimators [38, 41]. Clinical video systems target *binary* recognition [3, 23, 57]; in rodents, camera-only detectors have been built for PTZ-induced

mice [45], chronic-epilepsy screening [44], home-cage behavioral monitoring [21] and further mouse [26] and zebrafish [54] preparations, and Perlo *et al.* recently released RodEpile [42], the first public video-only rodent benchmark (13,053 clips, 19 rats) with a TimeSformer baseline at 97% F1 under subject-wise cross-validation. We validate our backbone on RodEpile directly and extend beyond binary detection to severity grading.

Multimodal video-EEG fusion is the clinical gold standard. Figure 4 shows the two streams on one clip from our cohort: the convulsive motion in the frames and the ictal discharge in the trace mark the same event, and an expert confirms a seizure only by reading them together. Computationally the pairing is rare, however: only a handful of human studies fuse both streams [1, 9, 31, 37], and in animals just two isolated attempts: late fusion of video, ECoG and piezo in rats [40], and cross-attention alignment of self-supervised EEG and video in mice [35]. Both address only *binary* detection; neither grades severity, the harder three- and five-class problem we study. We leave fusion to future work (Sec. 6); our aim here is to establish how far each single modality gets on its own.

3. Problem Setup

We now state the detection and grading problem formally, fixing the notation and evaluation protocol used throughout our experiments (Sec. 5). Figure 1 summarizes the pipeline.

3.1. Flat multi-class formulation

Clips and labels. A recording session provides temporally synchronized video and EEG. Around each expert-annotated event we extract a clip $c = (\mathbf{V}, \mathbf{E})$, where $\mathbf{V} \in \mathbb{R}^{T \times H \times W \times 3}$ is a short RGB video segment and $\mathbf{E} \in \mathbb{R}^{C \times L}$ the corresponding C -channel EEG trace. Matched non-seizure clips are sampled from seizure-free intervals of the same sessions. Each clip carries a severity label $y \in \{0, 1, 2, 3, 4\}$: 0 denotes non-seizure and 1–4 the ictal stages (Racine [43] Stage 2–5). Racine Stage 1 (mouth and facial automatisms) is annotated in this cohort but occurs in only six clips, too few to train or evaluate a class on, so we exclude it. The modeled label space begins at Stage 2 as a consequence of the data, and not because the scale starts there. The scale itself was derived from electrically kindled seizures, and its transfer to chemically induced models is not automatic [52]; we use the grades our annotator assigned and do not re-derive a model-specific scale. Each clip also belongs to a subject $s \in \mathcal{S}$ and a session $r \in \mathcal{R}$. The corpus pools clips from all subjects. Figure 2 shows a representative frame of the recording setup; because camera geometry differs per cage, each clip is cropped per cage before subjects are pooled.

Task granularities. We study three classification tasks of increasing difficulty, each obtained by collapsing the stage

label through a map $g_k : \{0, \dots, 4\} \rightarrow \{0, \dots, k-1\}$:

- **Detection** ($k=2$): $g_2(y) = \mathbf{1}[y > 0]$: seizure vs. non-seizure.
- **Coarse grading** ($k=3$): non-seizure / mild (Stage 2–3) / severe (Stage 4–5).
- **Fine grading** ($k=5$): the full non-seizure / Stage 2–5 scale.

Modalities. Each task pairs two *unimodal* predictors that we compare head-to-head: a video model $f_V : \mathbf{V} \mapsto \hat{y}$ and an EEG model $f_E : \mathbf{E} \mapsto \hat{y}$. The video model consumes a fixed number of sampled frames per clip. The EEG model is trained on short fixed windows and forms a clip decision by mean-pooling the window posteriors. We keep the two modalities *separate* by design: our goal is to measure how far each single modality gets on its own, so a fused predictor $f_{V,E}$ is out of scope and left to future work (Sec. 6).

Evaluation protocol. Seizure burden is highly uneven across subjects, so a single pooled score can mislead. We therefore adopt three safeguards throughout. (i) *Session-disjoint splits*: all clips from a session r fall entirely in train or validation, so no clip leaks across the split. The split seed is shared between the video and EEG models so their numbers are directly comparable. (ii) *Class imbalance* is handled with inverse-frequency-weighted cross-entropy in place of resampling. (iii) We report *macro-averaged* metrics (macro-F1 and, for detection, AUROC) together with the Matthews correlation coefficient (MCC); App. B.4 states each of them formally, along with the fold- and seed-aggregation convention. The two summarize different parts of the same confusion matrix: macro-F1 weights every stage equally, so the rare severe classes dominate it, while MCC is computed from the whole table and so follows the frequent classes. We report both so that no conclusion rests on one weighting, and always pair the pooled score with a per-subject breakdown, since aggregate accuracy is dominated by the few high-burden animals. Sec. 5 additionally reports a stricter protocol: held-out-subject generalization via subject-disjoint cross-validation.

3.2. Conditional ordinal severity grading

The $k=5$ task above places non-seizure and the four ictal stages on a single flat axis, which misstates the label space in two ways. First, “non-seizure” is not a point on the Racine scale [43]: the scale grades the motor manifestations of a seizure, so even its lowest rung is an ictal event, and the absence of an event is a different kind of object altogether. Second, a flat softmax treats the four stages as exchangeable, so predicting Stage 5 for a Stage 2 clip costs exactly what predicting Stage 3 costs, even though severity is ordered and the two errors differ in clinical consequence.

We therefore also study severity as a *conditional ordinal*

problem. Detection and grading factorize,

$$p(y | c) = \underbrace{p(\text{seizure} | c)}_{\text{detection}} \cdot \underbrace{p(s | c, \text{seizure})}_{\text{severity grading}}, \quad (1)$$

and we condition on the first factor: grading is posed on seizure clips alone, over the ordered stage set $s \in \{S2 \prec S3 \prec S4 \prec S5\}$. This matches how the annotation is produced, since an expert first confirms an event and then grades it, and it removes the 12,197 non-seizure clips that dominate the flat 5-class objective.

Ordinal head. Instead of four independent logits, an ordinal model predicts the $K-1$ cumulative decisions $p(s \succ S2)$, $p(s \succ S3)$, $p(s \succ S4)$, whose monotone structure encodes the ordering directly; the stage follows from how many thresholds are exceeded. The nominal four-way classifier trained on the same clips is the control, so any difference is attributable to the ordinal structure alone.

Metrics. Macro-F1 keeps continuity with the flat tasks, but it is order-blind, so we pair it with measures that are not: the mean absolute stage error $\frac{1}{N} \sum_i |\hat{s}_i - s_i|$, quadratic weighted κ , and the rate of *large-stage errors* $|\hat{s} - s| \geq 2$, which is the quantity a screening tool most needs to keep small. Because the deployment question is a new animal, we evaluate under the subject-disjoint five-fold protocol of Sec. 5.2.

4. Dataset

This study rests on a pooled twenty-subject synchronized video–EEG cohort. To our knowledge it is the largest paired rodent video–EEG corpus studied for learned seizure detection, and its scale, severity labels, and severely long-tailed burden are what make the head-to-head comparisons in Sec. 5 meaningful.

Acquisition. Synchronized video–EEG was recorded from twenty rats under chemically induced seizure protocols. Each animal was filmed in its home cage by a fixed overhead camera while implanted electrodes streamed synchronized EEG (Fig. 2). A domain expert annotated every seizure and graded its behavioral severity on the Racine scale [43]. Because camera geometry differs per cage, each clip is cropped per cage before subjects are pooled.

Clip extraction. Around each expert annotation we extract a seizure clip and sample a matched non-seizure clip from a seizure-free interval of the *same* session, yielding a pooled corpus of 24,131 clips over 601 recording sessions (12,197 non-seizure, 11,934 seizure); the same corpus expressed in recorded time is given in App. C.1, Table 12. Sampling the negative from the same session fixes camera, cage, and lighting across the pair, so a detector must separate the two from behavior alone (Fig. 4 shows one synchronized pair; the binary task is visualized in App. C.2, Fig. 10).

Severity stages. Each clip carries a Racine severity label. The behavioral signal grows more overt with stage, from

mild movement at Stage 2 to pronounced rearing and convulsive motion at Stage 4–5. This is the signal a video model reads directly and a single EEG channel cannot see, and it foreshadows the grading gap in Sec. 5. Per-stage *filmstrips* illustrating this progression are provided in the supplement (App. C.2, Fig. 11).

Long-tailed burden. Seizure burden is severely uneven across animals. The four busiest supply nearly half of all clips, subject totals span nearly three orders of magnitude (11–3,091 clips), and the most severe stage (S5, 193 clips in the entire cohort) stays rare throughout; the full per-subject composition is tabulated and visualized in the supplement (App. C.1, Table 13 and Fig. 9). This long tail is the central statistical difficulty of the cohort. A single pooled score is dominated by the few high-burden animals, so we report per-subject metrics alongside pooled ones and use inverse-frequency class weighting throughout. It is also why fine-grained severity grading is so hard.

Availability and de-identification. Raw recordings cannot be released, as they carry facility- and protocol-level identifiers. We instead release a fully de-identified cohort description: the per-subject statistics (App. C.1, Table 13), the task definitions, and the splitting protocol. All figures in this paper use de-identified frames with cage-side identifiers cropped or redacted. This makes the cohort’s composition and our protocol reproducible without exposing restricted data. The public RodEpi benchmark [42], on which we externally validate (Sec. 5), offers a fully open complement.

5. Experiments

We use the twenty-subject cohort of Sec. 4 to settle the question Sec. 1 left open: how far standard video understanding gets on automated rodent seizure detection and severity grading, and where it breaks. This is an applications-track study: we benchmark existing architectures under a controlled protocol; we do not propose a new algorithm.

Setup. Unless noted, the video model is an R(2+1)D-18 backbone [51] pretrained on Kinetics-400 [24] and fine-tuned on 16-frame 112×112 clips. As a matched EEG reference we train a GRU on the same cohort and split, echoing Ganguly *et al.* [18], and later broaden this to seven EEG models (Table 4). Splits are session-disjoint to prevent clip leakage, with the split seed shared between modalities so the two are directly comparable; we report macro-averaged metrics with per-subject breakdowns and inverse-frequency class weighting throughout (Sec. 3). Our headline comparison (Table 1) contrasts video and the EEG GRU across all three task granularities. Single runs use seed 42 throughout; Table 1 also reports macro-F1 as mean $\pm$ std over five seeds, which bounds how much of any margin below can be attributed to initialization. App. B describes every model in more detail, video backbones in App. B.1 and EEG models in App. B.2, together with the training recipe they all share

Table 1. Application results on the pooled twenty-subject cohort (session-disjoint). Video (R(2+1)D-18) vs. a matched EEG GRU. Precision, recall and F1 are macro-averaged; AUROC is binary at detection and macro one-vs-rest at grading; MCC is the multiclass Matthews correlation coefficient, computed from the full confusion matrix and so weighted towards the frequent classes that macro-F1 down-weights. Every entry is mean $\pm$ std over five seeds, so each comparison can be read against its own initialization noise; Tables 2, 4 and 3 instead report single seed-42 runs. EEG clip scores use logmean window pooling throughout, selected by the sweep of App. B.5. EEG edges video at detection on all five metrics—a 0.006 macro-F1 margin that is small but consistent (Welch $p=0.003$)—and video leads at both grading granularities. The one crossover is 5-class AUROC, where EEG ranks clips better (0.945 vs. 0.939) while scoring far worse once those rankings are turned into decisions (macro-F1 0.560 vs. 0.657). Best in **bold**, second-best underlined.

Task	Modality	Prec.	Rec.	F1	AUROC	MCC
2-class (detection)	Video	<u>0.968</u> ± 0.002	<u>0.968</u> ± 0.002	<u>0.968</u> ± 0.002	<u>0.993</u> ± 0.001	<u>0.936</u> ± 0.004
	EEG	0.975 ± 0.002	0.974 ± 0.003	0.974 ± 0.003	0.996 ± 0.001	0.949 ± 0.005
3-class (severity)	Video	0.775 ± 0.016	0.788 ± 0.009	0.780 ± 0.007	0.964 ± 0.003	0.853 ± 0.010
	EEG	<u>0.744</u> ± 0.014	<u>0.764</u> ± 0.013	<u>0.750</u> ± 0.010	<u>0.956</u> ± 0.002	<u>0.823</u> ± 0.014
5-class (severity)	Video	0.689 ± 0.024	0.643 ± 0.006	0.657 ± 0.009	<u>0.939</u> ± 0.004	0.831 ± 0.006
	EEG	<u>0.556</u> ± 0.040	<u>0.635</u> ± 0.035	<u>0.560</u> ± 0.005	0.941 ± 0.003	<u>0.728</u> ± 0.032

(App. B.3).

5.1. What model and input capacity is sufficient?

Before comparing modalities we ask what it takes to read seizure behavior from video at all. Four questions matter: how much signal the medium carries, how much network is needed to extract it, how much of each clip must be sampled, and how much of the result comes from shortcuts. We answer each in turn, holding the protocol of Sec. 3 fixed and varying one factor at a time.

Step 1: video carries a strong recognition signal. Fine-tuned on 16-frame 112×112 clips, R(2+1)D-18 reaches 0.968 ± 0.002 macro-F1 and 0.993 ± 0.001 AUROC at detection over five seeds (Table 1). The clip construction makes this a test of behavior rather than of scene: every seizure clip is paired with a non-seizure clip from the *same* animal and the *same* session, so camera, cage and lighting are identical within a pair and cannot by themselves separate the two (App. C.2, Fig. 10). The result holds animal by animal. Per-subject detection macro-F1 exceeds 0.90 on the well-represented subjects, 0.974 on R1 and 0.906 on R7, and degrades gracefully on the near-silent ones, where only a handful of validation clips exist. Behavior alone therefore supports near-ceiling detection on this cohort, echoing the false-positive reductions video brought to human vEpiNet [31]. Sec. 5.2 takes up how it compares against an electrode-based reference.

Step 2: architecture capacity is second-order. If the signal is that strong, how much network does reading it re-

quire? We fine-tune ten video networks under a matched 12-epoch budget on the same cohort, split and task (Table 2). They span factorized and separable convolution, efficient and two-pathway convolution, hierarchical and multiscale attention, and convolution-free attention with self-supervised pretraining; App. B.1 describes each. All ten land within a 0.03 macro-F1 band, so detection is essentially *backbone-agnostic*, and the efficient X3D-M matches R(2+1)D with 4M parameters against 33M (Fig. 3). The ordering shifts with the data regime and the clustering persists at grading; App. E.3 gives the per-model read-out.

To check that this comparison is not simply reading initialization noise, we replicated every cell of Table 2 over five seeds. The spread across backbones is large relative to the spread within one: at detection the ten span a 0.027 macro-F1 band against a median per-backbone seed deviation of 0.002, a ratio of 14, rising to 8 at 3-class (band 0.065, median deviation 0.009) and 10 at 5-class (band 0.089, 0.009). The ranking is therefore real, and *backbone-agnostic* should be read as a statement about the narrowness of the band rather than about the ordering being arbitrary. Two cells deserve their own note. X3D-M is the strongest model at 3-class (0.794 ± 0.005) and the weakest at 5-class (0.586 ± 0.003); both deviations are small, so the reversal is a stable property of the architecture on this task and not an unlucky run. VideoMAE-B is the least stable model by a wide margin, at ± 0.030 macro-F1 at 3-class against a median of 0.009, which is the one place where a single-seed reading of Table 2 could mislead. Per-backbone means and deviations are in Table 10. Two further results point the same way: Kinetics pretraining is worth only about 0.02 macro-F1 over random initialization, and detection saturates well before the full cohort is used (App. E.1). Neither model capacity nor data volume is the binding constraint here.

Step 3: sufficient temporal and spatial sampling. How much of each clip must the model actually see? A single frame recovers little: macro-F1 falls to 0.569, against 0.970 at 16 frames (Table 3). The model still drives its training loss to near zero, so the limit is the evidence in one frame, not the optimizer. The matched-negative design makes this a clean control. A positive and its negative share rat, cage, camera and lighting, so a still frame carries almost no usable cue (App. C.2, Fig. 10). The collapse holds at every granularity: one frame reaches 0.417 macro-F1 at 3-class and 0.314 at 5-class. Appearance alone carries neither the event nor its severity. Detection on this cohort is a genuinely *temporal* problem: the discriminative evidence lives in how the animal moves.

Accuracy then rises steeply to eight frames and peaks at 32 at every granularity, while 64 returns nothing and costs 0.035 macro-F1 at 5-class. Spatial resolution runs the other way: accuracy falls monotonically from 112 to 448 pixels (App. E.3). The clips are only ~ 300 –480 px native, so larger

inputs upsample the recording and spend compute without adding detail. We therefore keep 16 frames at 112 px throughout, a budget that concedes almost nothing to 32 frames. App. E.3 gives the full read-out and the comparison against seed noise.

Step 4: how much of this could be a shortcut? A detector can reach high accuracy on the wrong cue. Three structural arguments bound that risk here. Negatives are drawn from the same session as their positive, so camera pose, cage, bedding and illumination are fixed within a pair and carry no label information (App. C.2, Fig. 10). Clips are cropped per cage before pooling, so inter-animal framing cannot identify an animal and, through it, its seizure burden. And the single-frame control of Step 3 collapses to 0.569, so a static view of the scene supplies little. Grad-CAM [47] maps concentrate on the animal during convulsive frames and stay diffuse on non-seizure clips, across Stages 2, 3 and 5 (Fig. 5). That is a partial answer to the artifact-disambiguation question fusion methods raise [25]. We read it as consistent with a behavioral detector, though short of proof. The controls exclude particular shortcuts, Grad-CAM only shows where the model looks, and the interventions that would settle the question are unrun. App. D.3 sets out what this evidence does and does not establish.

Key takeaway. Detection is limited by temporal context rather than by model size. Ten backbones spanning every paradigm land within 0.03 macro-F1 and a 4M-parameter model ties a 33M one, while a single frame collapses to 0.569. Sixteen to thirty-two frames at 112 px suffices, and the shortcut controls available to us are consistent with a behavioral detector without settling the question.

5.2. Cross-modal comparison, generalization and external validation

With the capacity question settled, we turn to how far the video result carries: how it stands against an electrode-based reference, whether it survives held-out animals and harder label granularities, and whether it reproduces on a public benchmark.

Is the EEG baseline strong enough? Because the headline is a cross-modal comparison, the EEG side must be a strong baseline. Each EEG model consumes the first recorded EEG channel decimated to 125 Hz, cut into 6 s windows at 3 s stride and z-scored per window; a clip decision is the log-mean of the per-window posteriors (App. B.5). On these identical windows, split and seed we evaluate seven EEG models (Table 4), chosen to span how the temporal structure of a trace can be modeled. Five are deep. The GRU above and an LSTM [20] read the window as a raw sequence. EEGNet [29] is a compact CNN whose first layer acts as a learned bandpass filter bank, the EEG Conformer [49] puts a self-attention encoder over convolutional tokens, and a dilated temporal convolutional network

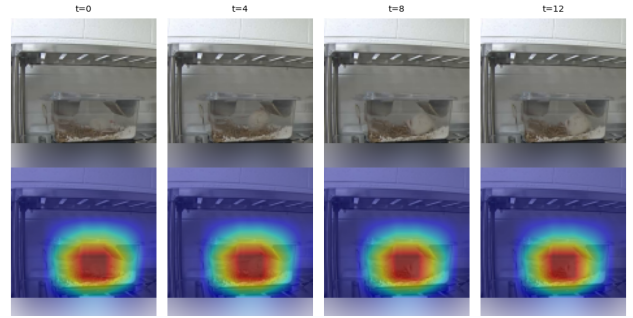
Table 2. Video backbone comparison (identical split, seed and 12-epoch budget). Every entry is mean $\pm$ std over five seeds (1, 2, 3, 5, 42), the deviation set as a subscript; $n=5$ except SlowFast detection and TimeSformer detection AUROC, which are $n=4$ (one SlowFast cell’s stored predictions were overwritten and are excluded; the TimeSformer seed-42 run predates the patch that saved per-clip predictions, so its F1 survives but its AUROC cannot be recomputed). Precision/recall/F1 are macro-averaged; AUROC is binary at detection and macro one-vs-rest at grading; MCC is the multiclass Matthews correlation coefficient. All ten architectures fall within a 0.03 macro-F1 band at detection and stay clustered at grading, and MCC reproduces that clustering. Point values are the seed-42 run, matching Tables 4 and 3; every cell was additionally replicated over five seeds, and the between-backbone spread exceeds the median within-backbone seed deviation by $14\times$, $8\times$ and $10\times$ at the three granularities respectively. Table 10 gives the per-backbone five-seed means. Paradigms and parameter counts are in App. B.1, Table 9, in the same row order. Best per column **bold**, second-best underlined.

Backbone	Detection (2-class)					3-class grading					5-class grading				
	P	R	F1	AUC	MCC	P	R	F1	AUC	MCC	P	R	F1	AUC	MCC
MViTV2-S [30]	<u>0.972</u> ± 0.001	<u>0.972</u> ± 0.001	<u>0.972</u> ± 0.001	0.993 ± 0.002	<u>0.944</u> ± 0.003	0.780 ± 0.014	0.800 ± 0.027	0.786 ± 0.005	0.971 ± 0.002	0.856 ± 0.017	0.694 ± 0.032	0.656 ± 0.030	0.660 ± 0.016	0.915 ± 0.011	0.832 ± 0.007
SlowFast-R50 [17]	0.974 ± 0.002	0.974 ± 0.002	0.974 ± 0.002	0.995 ± 0.001	0.948 ± 0.004	<u>0.781</u> ± 0.018	<u>0.804</u> ± 0.013	<u>0.789</u> ± 0.009	<u>0.966</u> ± 0.001	<u>0.859</u> ± 0.012	<u>0.718</u> ± 0.024	<u>0.655</u> ± 0.014	<u>0.675</u> ± 0.008	<u>0.939</u> ± 0.011	0.851 ± 0.003
R(2+1)D-18 [51]	0.971 ± 0.001	0.971 ± 0.001	0.971 ± 0.001	<u>0.994</u> ± 0.001	0.942 ± 0.002	0.780 ± 0.007	0.787 ± 0.007	0.783 ± 0.002	0.960 ± 0.003	0.860 ± 0.003	0.734 ± 0.027	0.643 ± 0.011	<u>0.667</u> ± 0.010	0.928 ± 0.017	<u>0.841</u> ± 0.007
X3D-M [16]	0.969 ± 0.002	0.969 ± 0.002	0.969 ± 0.002	0.993 ± 0.001	0.937 ± 0.003	0.782 ± 0.008	0.810 ± 0.015	0.794 ± 0.005	0.952 ± 0.005	0.856 ± 0.008	0.568 ± 0.004	0.616 ± 0.013	0.586 ± 0.003	0.911 ± 0.003	0.824 ± 0.007
MViTV1-B [15]	0.969 ± 0.002	0.969 ± 0.002	0.969 ± 0.002	0.991 ± 0.003	0.938 ± 0.003	0.771 ± 0.018	0.794 ± 0.016	0.779 ± 0.011	0.965 ± 0.003	0.849 ± 0.016	0.687 ± 0.022	0.643 ± 0.011	0.653 ± 0.008	0.935 ± 0.012	0.836 ± 0.006
Video Swin-S [34]	0.966 ± 0.003	0.966 ± 0.003	0.966 ± 0.003	0.987 ± 0.004	0.932 ± 0.005	0.767 ± 0.013	0.780 ± 0.022	0.771 ± 0.009	0.962 ± 0.011	0.834 ± 0.016	0.716 ± 0.032	0.634 ± 0.017	0.645 ± 0.011	0.916 ± 0.025	0.818 ± 0.012
S3D [55]	0.967 ± 0.002	0.967 ± 0.002	0.967 ± 0.002	0.992 ± 0.001	0.933 ± 0.005	0.756 ± 0.014	0.782 ± 0.027	0.764 ± 0.004	0.958 ± 0.006	0.840 ± 0.015	0.644 ± 0.015	0.639 ± 0.021	0.636 ± 0.009	0.917 ± 0.008	0.813 ± 0.015
Video Swin-T [34]	0.963 ± 0.001	0.962 ± 0.001	0.963 ± 0.001	0.990 ± 0.003	0.925 ± 0.002	0.761 ± 0.014	0.776 ± 0.020	0.766 ± 0.008	0.963 ± 0.002	0.829 ± 0.006	0.660 ± 0.022	0.639 ± 0.009	0.632 ± 0.002	0.943 ± 0.009	0.796 ± 0.006
VideoMAE-B [50]	0.958 ± 0.004	0.958 ± 0.004	0.958 ± 0.004	0.986 ± 0.002	0.915 ± 0.007	0.729 ± 0.033	0.736 ± 0.042	0.729 ± 0.030	0.942 ± 0.018	0.771 ± 0.063	0.704 ± 0.018	0.623 ± 0.025	0.636 ± 0.017	0.920 ± 0.013	0.802 ± 0.013
TimeSformer [7]	0.945 ± 0.001	0.944 ± 0.001	0.945 ± 0.001	0.984 ± 0.003	0.889 ± 0.002	0.722 ± 0.017	0.761 ± 0.017	0.735 ± 0.013	0.950 ± 0.002	0.778 ± 0.019	0.650 ± 0.014	0.594 ± 0.014	0.601 ± 0.009	0.913 ± 0.010	0.757 ± 0.028

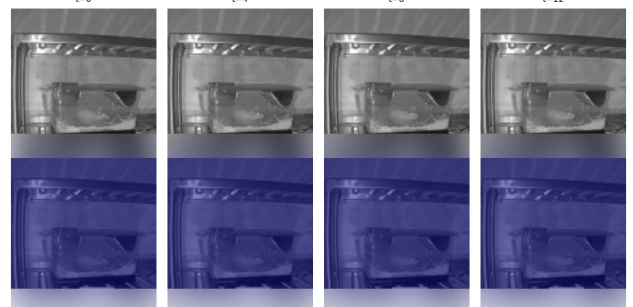
Table 3. Temporal-sampling ablation: R(2+1)D macro-F1 / AUROC against frames per clip, at all three granularities (grading AUROC is macro one-vs-rest; MCC is the multiclass Matthews correlation coefficient; batch size scales with frame count). A single frame recovers little at any granularity, so the task rests on motion. Every granularity peaks at 32 frames; 64 returns nothing and costs 0.035 macro-F1 at 5-class. Best per column **bold**, second-best underlined.

Frames	Detection			3-class			5-class		
	F1	AUC	MCC	F1	AUC	MCC	F1	AUC	MCC
1	0.569	0.615	0.149	0.417	0.599	0.152	0.314	0.634	0.147
4	0.940	0.983	0.881	0.739	0.950	0.760	0.612	0.926	0.741
8	0.959	0.989	0.918	0.763	0.957	0.838	0.627	0.919	0.789
16	0.970	0.993	0.941	0.780	0.962	0.857	0.662	0.946	0.830
32	0.972	0.994	0.943	0.793	0.971	0.858	0.667	0.951	0.835
64	<u>0.971</u>	<u>0.994</u>	<u>0.941</u>	<u>0.785</u>	<u>0.971</u>	0.848	0.632	<u>0.950</u>	0.807

(TCN) [5] grows its receptive field exponentially with depth. Two are classical: Random Forest [8] and XGBoost [12] learn no representation and instead consume a fixed per-window descriptor of band powers and time-domain statistics. The strongest at detection is the TCN, at 0.978 ± 0.001 macro-F1 and 0.996 AUROC, a little above the GRU (0.974 ± 0.003); it also leads at 3-class (0.767 vs. 0.750). At 5-class the two are tied within seed noise (0.563 ± 0.012 vs. 0.560 ± 0.005), so the ordering that holds across the table is $\text{TCN} \geq \text{GRU} > \text{Conformer} > \text{LSTM} > \text{EEGNet}$ rather than a single winner. The spread across the five deep models is 0.055 macro-F1 at detection but 0.147 at 3-class and 0.154 at 5-class: architecture is close to irrelevant for detecting a seizure and matters considerably for grading one, the mirror image of the backbone-agnostic result on the video side (Table 2). The classical baselines are competitive at detection, where feature-based XGBoost reaches 0.914 against



(a) Seizure clip (Stage 4): activation concentrates on the animal



(b) Non-seizure clip: activation is diffuse, with no focus on the animal

Figure 5. Grad-CAM [47] for the R(2+1)D detector; each strip shows input frames (top) and seizure-class activation (bottom), for four of the eight sampled frames. (a) On a Stage 4 clip, activation concentrates on the animal during convulsive motion. (b) On a non-seizure clip it stays diffuse. The contrast indicates the detector responds to behavior rather than cage background. All eight frames at full width are in App. D.2, Fig. 14. De-identified clips; identifier bands redacted.

the deep EEGNet’s 0.923, but fall away at grading (0.466 vs. 0.563 for the TCN at 5-class), so the deep advantage on EEG

is small where the task is easy and real where it is hard. The LSTM is the least reliable model in the field: two of its fifteen runs failed to converge outright, which no other backbone did. One caveat bounds the absolute margin: this EEG reference is single-channel, so a richer montage could narrow the reported cross-modal gap.

Architecture diversity is not tuning, so we also sweep the EEG side’s window, stride, sampling rate and clip-pooling rule, and every model in Table 4 is trained at the winning setting (App. B.5). The sweep is informative about which axis matters. Window length is nearly flat—2, 4, 6 and 8 s span 0.015 macro-F1, less than the 0.018 range we measure across seeds at a fixed setting—while stride is monotone (0.554 at 1.5 s to 0.496 at 6 s) and sampling rate is the largest single effect (dropping to 62.5 Hz costs 0.049). The clip-pooling rule, an axis rarely swept at all, is worth more than any of them: log-mean beats the usual mean on seven of eight configurations and gains 0.028 macro-F1 at the winning setting, for free, since it re-scores stored window posteriors without retraining. Because pooling and window interact—the best window under mean pooling is only fourth-best under log-mean—we select the best *pair* rather than optimizing the axes in sequence. Even so, the tuned EEG reference does not close the cross-modal gap: the strongest EEG model at 5-class reaches 0.563 against video’s 0.657.

Does the ordering depend on the metric? Macro-F1 and MCC weight the confusion matrix differently, so we report both throughout (Sec. 3). They disagree about how hard grading *looks*: at 5-class video scores 0.657 macro-F1 against 0.831 MCC. They agree on every ordering that carries a claim. The cross-modal gap widens with granularity under both, and the ten backbones stay clustered under both. The headline conclusions are therefore a property of the data, not of the summary statistic; App. B.4 works this through.

Generalization to unseen subjects. Session-disjoint validation still trains and tests on the *same* animals. The stricter, deployment-relevant question is whether the detector transfers to an animal it has never seen, the protocol RodEpil [42] adopts. We therefore repeat all three tasks under *subject-disjoint* five-fold cross-validation, holding out four whole animals per fold (Table 5). At detection, neither modality is much troubled by the harder protocol. Video degrades only marginally, from 0.968 to 0.948 macro-F1 and 0.993 to 0.985 AUROC. The EEG GRU is essentially unchanged, moving from 0.974 to 0.972, and the two fold spreads are nearly identical (σ 0.016 against 0.019). The two are therefore equally stable across held-out animals, and EEG ends up marginally ahead. That reverses the session-disjoint ordering, though both differences are small relative to the fold spread (Fig. 6). Detection generalizes across animals in both modalities; the modalities separate at grading, not here. Part

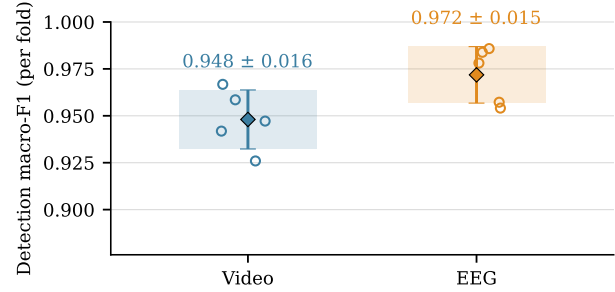


Figure 6. Cross-subject stability under subject-disjoint 5-fold CV (per-fold detection macro-F1 from Table 5; dots are folds, diamonds are the mean, whiskers/bands are $\pm 1\sigma$). Both modalities are tightly and comparably clustered, video at 0.948 ± 0.016 over [0.926, 0.967] and EEG at 0.972 ± 0.015 over [0.954, 0.986], so detection transfers to unseen animals about equally well either way, with EEG marginally ahead on the mean. The modalities diverge at grading instead (Table 5).

of the residual cross-subject difficulty is visual: cage framing, lighting, and coat vary from animal to animal (App. C.3, Fig. 12).

Grading is where the protocol bites. Holding out whole animals costs video 0.060 macro-F1 at 3-class and 0.182 at 5-class ($0.780 \rightarrow 0.720$ and $0.657 \rightarrow 0.475$), against 0.094 and 0.156 for the EEG reference, so the cross-modal margin at 5-class narrows from 0.097 to 0.071. Video still leads at both grading granularities: 0.720 against 0.656 at 3-class, and 0.475 against 0.404 at 5-class. The two now degrade at comparable rates, and video is no longer uniformly the steadier across folds. We read the larger video drop as the price of appearance: a held-out animal arrives with unfamiliar cage framing, lighting and coat, and at five classes the model must separate stages that differ by subtle motion on top of that shift. Fine-grained grading across unseen animals is thus the weakest regime we measure for either modality, and the accurate statement of our result is that video’s advantage is largest within a cohort and narrower, though still substantial, across new animals.

Grading: video degrades, EEG collapses. Severity is harder, and here the two modalities part ways. As the task moves from 2-class detection to 3-class (non-seizure / mild / severe) and 5-class (non-seizure / Stage 2–5) grading, video macro-F1 declines gracefully from 0.968 to 0.780 to 0.657 and stays usable even at five levels. Even the rare severe classes retain a workable signal (Stage 5 per-class F1 0.42 at 5-class; App. E.3, Table 17). The EEG reference declines more steeply from the same starting point ($0.974 \rightarrow 0.750 \rightarrow 0.560$), and its rarest classes are where it gives way (Stage 4 F1 0.20, Stage 5 F1 0.12 against 0.36 and above for video; the per-class breakdown is given in Table 17 and Figure 8). Video thus retains a workable severity signal where EEG does not, though fine-grained grading re-

Table 4. EEG backbone comparison (identical windows and split) across all three granularities, as mean $\pm$ std over five seeds. Precision/recall/F1 are macro-averaged; detection AUROC is binary, grading AUROC macro one-vs-rest; MCC is the multiclass Matthews correlation coefficient. All models use the configuration selected in App. B.5: 6 s windows at 3 s stride, 125 Hz, logmean clip pooling. The TCN leads at detection and 3-class; at 5-class it and the GRU are tied within noise (0.563 ± 0.012 vs. 0.560 ± 0.005). Random Forest and XGBoost use per-window band-power and time-domain features; XGBoost is deterministic given the data, hence ± 0.000 . Two of fifteen LSTM runs failed to converge (macro-F1 0.231 and 0.136 against cell medians of 0.677 and 0.456); they are excluded from the LSTM means, which therefore rest on $n=3$ at 3-class and $n=4$ at 5-class. Best per column **bold**, second-best underlined.

EEG backbone	Detection (2-class)					3-class					5-class				
	P	R	F1	AUC	MCC	P	R	F1	AUC	MCC	P	R	F1	AUC	MCC
<i>Deep</i>															
GRU [18]	<u>0.975</u> ± 0.003	<u>0.974</u> ± 0.003	<u>0.974</u> ± 0.003	<u>0.996</u> ± 0.001	<u>0.949</u> ± 0.005	<u>0.744</u> ± 0.014	0.764 ± 0.013	<u>0.750</u> ± 0.010	<u>0.956</u> ± 0.002	<u>0.823</u> ± 0.014	0.556 ± 0.040	<u>0.635</u> ± 0.035	<u>0.560</u> ± 0.005	0.941 ± 0.003	0.728 ± 0.032
LSTM [20]	0.972 ± 0.008	0.971 ± 0.007	0.971 ± 0.008	0.994 ± 0.004	0.942 ± 0.015	0.696 ± 0.041	0.741 ± 0.067	0.698 ± 0.053	0.937 ± 0.027	0.743 ± 0.059	0.476 ± 0.042	0.555 ± 0.100	0.468 ± 0.052	0.903 ± 0.039	0.598 ± 0.059
EEGNet [29]	0.926 ± 0.011	0.922 ± 0.013	0.923 ± 0.012	0.975 ± 0.005	0.848 ± 0.023	0.646 ± 0.010	0.676 ± 0.018	0.620 ± 0.015	0.903 ± 0.005	0.626 ± 0.031	0.448 ± 0.040	0.488 ± 0.025	0.409 ± 0.039	0.864 ± 0.011	0.499 ± 0.037
EEG Conformer [49]	0.972 ± 0.001	0.972 ± 0.001	0.972 ± 0.001	0.988 ± 0.002	0.943 ± 0.001	0.719 ± 0.009	<u>0.784</u> ± 0.013	0.729 ± 0.010	0.952 ± 0.002	0.780 ± 0.017	0.512 ± 0.004	0.624 ± 0.032	0.523 ± 0.005	0.936 ± 0.008	0.683 ± 0.008
TCN [5]	0.979 ± 0.001	0.978 ± 0.001	0.978 ± 0.001	0.996 ± 0.001	0.957 ± 0.002	0.758 ± 0.014	0.786 ± 0.008	0.767 ± 0.007	0.960 ± 0.001	0.839 ± 0.015	<u>0.537</u> ± 0.008	0.654 ± 0.013	0.563 ± 0.012	<u>0.941</u> ± 0.003	<u>0.728</u> ± 0.021
<i>Classical (band-power + statistical features)</i>															
Random Forest [8]	0.885 ± 0.001	0.852 ± 0.001	0.852 ± 0.001	0.986 ± 0.000	0.737 ± 0.002	0.541 ± 0.001	0.542 ± 0.001	0.529 ± 0.001	0.934 ± 0.001	0.613 ± 0.002	0.477 ± 0.001	0.328 ± 0.001	0.332 ± 0.002	0.916 ± 0.002	0.503 ± 0.002
XGBoost [12]	0.922 ± 0.000	0.913 ± 0.000	0.914 ± 0.000	0.984 ± 0.000	0.835 ± 0.000	0.657 ± 0.000	0.684 ± 0.000	0.642 ± 0.000	0.934 ± 0.000	0.623 ± 0.000	0.479 ± 0.000	0.530 ± 0.000	0.466 ± 0.000	0.924 ± 0.000	0.529 ± 0.000

Table 5. Generalization from session-disjoint to subject-disjoint (5-fold, whole animals held out) at all three granularities. The protocols separate the two regimes cleanly. At detection, holding out whole animals costs video 0.022 macro-F1 and costs the EEG reference nothing ($0.974 \rightarrow 0.972$), so EEG is ahead on unseen animals and the two are equally stable across folds (± 0.015 vs. ± 0.016 ; MCC ± 0.030 vs. ± 0.031). At grading video leads under both protocols, though it gives up more when animals are held out, 0.187 macro-F1 at 5-class against 0.159 for EEG, so the margin narrows from 0.128 to 0.100. Subject-disjoint entries are mean $\pm$ std over folds.

Task	Mod.	Session-dis.		Subject-disjoint (5-fold)		
		Macro-F1	MCC	Macro-F1	AUROC	MCC
Detection	Video	0.968	0.936	0.948 ± 0.016	0.985 ± 0.006	0.896 ± 0.031
	EEG	0.974	0.949	0.972 ± 0.015	0.995 ± 0.005	0.944 ± 0.030
3-class	Video	0.780	0.853	0.720 ± 0.028	0.942 ± 0.015	0.761 ± 0.029
	EEG	0.750	0.823	0.656 ± 0.022	0.916 ± 0.025	0.793 ± 0.046
5-class	Video	0.657	0.831	0.475 ± 0.024	0.888 ± 0.026	0.676 ± 0.041
	EEG	0.560	0.728	0.404 ± 0.042	0.831 ± 0.076	0.657 ± 0.107

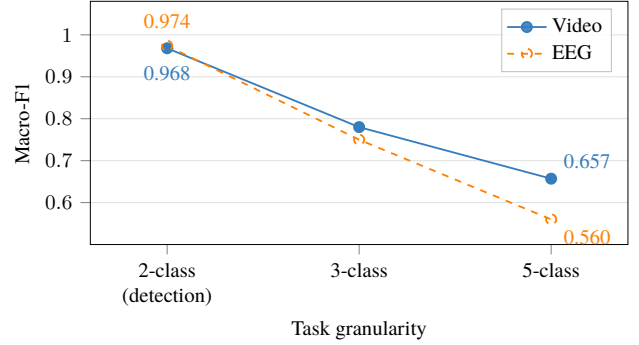


Figure 7. Video degrades more gracefully than EEG as the task hardens (macro-F1, values from Table 1). The two modalities are level at detection (0.968 vs. 0.974), and by 5-class grading video still retains a usable 0.657 whereas the EEG GRU falls to 0.560, so a 0.006 EEG lead at detection becomes a 0.097 video lead at five classes.

mains the weakest regime for both. The hardest cases are rare mild events and data-poor animals (App. D.1, Fig. 13). Figure 7 traces the two curves side by side: they start level but the video line stays high while the EEG line dives, the gap widening from 0.004 at detection to 0.128 at 5-class.

Do the two modalities fail the same way? An aggregate gap says which model is better, not whether the other is redundant, so we compare the errors themselves (App. E.4, Fig. 16). They differ in kind. Video’s mistakes are ordinal near-misses: 72.4% land on an adjacent stage, and the largest single cell is Stage 4 read as Stage 3 (95 of 167 clips). It registers that a convulsion is underway and misjudges the degree. The EEG reference adds two failure modes video does not have. It loses events outright, calling 10.1% of

Stage 3 clips non-seizure against 2.3% for video, and its errors ignore the ordinal scale, 44% landing two or more stages away against 28% for video (mean $|\Delta_{\text{stage}}|$ 1.51 vs. 1.30). A single channel supports a seizure/no-seizure decision without carrying a monotone severity axis. The one exception is the informative one: at Stage 2, the rung with the least whole-body movement, the ordering reverses and EEG recalls 0.863 against video’s 0.650. Video’s advantage is behavioral, and it recedes exactly where the behavior does. Across the clips both models scored, video alone is correct on 14.2% and EEG alone on 4.9% (Table 18), so each modality recovers clips the other loses. That is a measured case for the fusion we leave to future work (Sec. 6).

External validation on RodEpilep. To test whether these detection numbers transfer beyond our own cohort, we evaluate the same R(2+1)D backbone on the public RodEpilep benchmark [42] (13,053 clips, 19 rats) under its native *subject-*

Table 6. External validation on RodEpiL [42] (subject-wise 5-fold, seizure-class F1 and MCC). The raw release is separable by an encoding artifact that inflates detection to near-perfect; re-encoding every clip through one uniform pipeline removes the shortcut and yields honest scores on par with the dataset’s TimeSformer baseline (0.97).

	F0	F1	F2	F3	F4	Mean
Raw (leaked) F1	1.000	0.998	1.000	1.000	1.000	0.9996
Raw (leaked) MCC	1.000	0.998	1.000	1.000	1.000	0.9995
Re-encoded F1	0.965	0.978	0.997	0.977	0.979	0.979
Re-encoded AUROC	0.999	1.000	1.000	1.000	0.998	0.999
Re-encoded MCC	0.957	0.976	0.996	0.967	0.964	0.972

wise five-fold protocol. A direct run scores a suspicious 0.9996 F1, because the raw release leaks the class label into low-level clip encoding. Every normal clip is stored at 30 fps / 301 frames and every seizure clip at 29 fps / 291 frames. Either attribute alone separates the classes with AUROC 1.0, and the per-clip crop resolution also differs by class. Re-encoding all clips through a single uniform pipeline (224×224 letterbox, 30 fps, 300 frames, identical codec) removes this shortcut, and the backbone then reaches **0.979 subject-wise F1** (AUROC 0.999; per-fold in Table 6). This matches the dataset’s TimeSformer baseline (0.97) and reflects genuine behavior: the per-fold learning curve now rises gradually instead of saturating at epoch one. We flag the encoding confound as a caution for future users of this otherwise valuable benchmark.

All backbones on RodEpiL. Running the full ten-backbone suite on re-encoded RodEpiL reshuffles the ranking and widens it, from a 0.03 band on our own cohort to 0.87–0.99 macro-F1. The convolutional networks lead and the attention models trail, the reverse of our ordering. Backbone choice is therefore a data-regime effect, not a universal ranking (App. E, Table 15).

Key takeaway. Video matches the EEG reference at detection (0.968 against 0.974) and pulls ahead as the task becomes behavioral, reaching 0.657 against 0.560 at five-class. That ordering survives both held-out animals and the public RodEpiL benchmark, though the margin is widest within a cohort and narrows once whole animals are held out. The two also fail differently: video misgrades by one stage, EEG loses events and scatters across non-adjacent ones, and at Stage 2 the ordering reverses. Their errors are complementary.

5.3. Does inverse-frequency weighting keep the long tail from dominating the result?

Two independent imbalances run through this cohort, and both threaten to turn a pooled score into a report on the majority. The first is across stages: the five-class validation set is 51% non-seizure and 41% Stage 3, while Stage 5 is 34

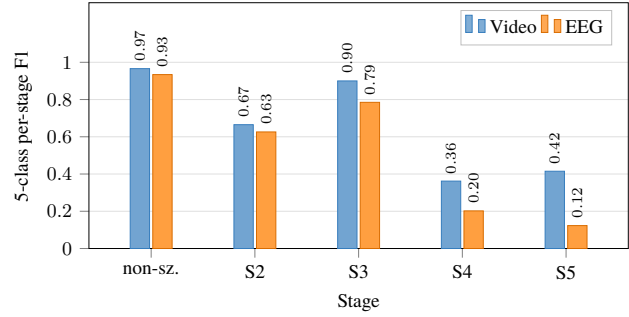


Figure 8. Per-stage F1 at the hardest 5-class grading task (Table 17). Both modalities are strong on the common classes, non-seizure and Stage 3. On the rare severe stages EEG gives way, with S4/S5 F1 of 0.20 and 0.12, while video retains a usable signal at 0.36 and 0.42. Video holds a workable severity cue where the electrode-based model gives out.

clips, a ratio of 80:1 between the largest and smallest class. The second is across animals, whose totals span 11 to 3,091 clips (App. C.1). A model that ignored every rare stage and every quiet animal could still look strong on aggregate accuracy.

Three defenses, at construction, objective and measurement. Detection sidesteps the problem by construction: each seizure clip is paired with a matched negative, so the binary task is near-balanced (12,197 against 11,934) and imbalance enters only at grading. At grading the objective carries it: we train with inverse-frequency-weighted cross-entropy, weights normalized to mean one, which at five-class gives Stage 5 roughly $60\times$ the weight of non-seizure. Measurement then keeps the tail visible. Every score is macro-averaged, every pooled number is paired with a per-subject breakdown, and the generalization study holds out whole animals (Table 5). A model that quietly abandons the tail cannot hide behind a pooled figure. One caveat applies to the metrics themselves. AUROC is optimistic under strong imbalance [46], which is why macro-F1 carries the argument at the rare stages.

The tail stays learnable. The defenses work in the sense that matters first: the rare stages are predicted rather than weighted out of existence. A majority-class predictor reaches only 0.136 macro-F1 at five-class, against the video model’s 0.657, and the model still recovers a third of the Stage 5 clips (recall 0.324 on 34 examples) at 0.579 precision (Table 17).

What it costs. Re-weighting redistributes error rather than removing it, and two limits follow. The tail remains the weakest regime: Stage 4 and Stage 5 F1, 0.362 and 0.415, sit far below Stage 3 at 0.900. The correction mitigates imbalance without solving it. More instructive is what happens when the same correction meets a weaker signal. The EEG model’s Stage 5 precision is 0.014 against recall 0.121: it

assigns Stage 5 to roughly nine times as many clips as actually carry it, converting a rare-class blind spot into a false-positive one. Up-weighting a class the modality cannot separate buys recall at a price in precision that a screening tool would not accept. The imbalance strategy should therefore be chosen per modality, not globally.

One caveat bounds all of this. We report the behavior of a single reasonable strategy: inverse-frequency weighting is never compared against unweighted training, resampling [11] or a focal objective [32], three standard options surveyed in [22] (Sec. 6).

Key takeaway. Weighting keeps the tail learnable rather than solving it. Rare stages are predicted instead of collapsing to the majority (0.657 macro-F1 against 0.136 for a majority-class predictor, with Stage 5 recall 0.324 on 34 clips), yet they remain the weakest regime, and on the weaker modality the same correction over-predicts Stage 5 roughly ninefold.

5.4. Why behavioral video matters

Taken together, the results show a camera matching electrodes at detection under both protocols and pulling ahead once the endpoint is itself behavioral. Three consequences follow.

Video reads the endpoint directly. The Racine scale grades motor manifestations: twitching, rearing, forelimb clonus, falling [43]. A camera observes the signal the grade is defined on, while an electrode observes a correlate of it. That is the simplest account of Fig. 7. The modalities are indistinguishable while the question is whether an event occurred, and separate once it becomes how severe that event was, with the largest gap at the rarest and most severe stage (Stage 5 per-class F1 0.42 against 0.12).

Cost, scale and welfare favor the camera. Electrode EEG needs surgical implantation, which is expensive, low-throughput, and a source of infection, signal drift and welfare burden. A camera is cheap, surgery-free, and watches many cages at once, so video-only screening scales to the cohorts preclinical drug discovery needs and reduces harm in the spirit of the 3Rs. Deployment is equally favorable: the 4M-parameter X3D-M matches the 33M R(2+1)D (Table 9), which puts continuous monitoring on modest hardware within reach.

Electrodes remain necessary, and fusion is the next step. Our EEG reference is a single channel, and its limits show throughout: it loses 10.1% of Stage 3 events outright, carries no monotone severity axis, and supplies no localization. A richer montage would ease all three. Even so, EEG captures electrographic-only seizures that have no motor correlate at all, recovers onset timing, and recovers the Stage 2 events video misses (Sec. 5.2); a camera is blind to each of these. Video does not displace the electrode; it carries the behavioral half of the workflow, which is the half that consumes

expert time. We keep the two streams unfused here deliberately. A fused model would confound the question we set out to answer, namely how far each modality gets on its own. No shared paired video–EEG benchmark yet exists on which such a model could be compared against prior work (Sec. 6).

Key takeaway. A camera alone matches electrode-based detection, on held-out sessions and held-out animals alike, and grades severity more accurately. Electrodes stay necessary for electrographic-only events, onset timing, and the mildest behavioral stage.

6. Limitations and Future Work

Our study is deliberately scoped to unimodal models on one preclinical cohort, and five limitations follow from that scope. Generalization rests on five subject-disjoint folds of a single cohort, so multi-cohort, subject-wise evaluation remains the honest test, and fine-grained severity on unseen animals is where video gives up the most (0.662 to 0.475 macro-F1, Table 5). Severity grading is the open task: the rare severe stages carry most of the error and invite targeted collection or the ordinal formulation of Sec. 3.2, and our inverse-frequency weighting is reported without an ablation against resampling [11] or a focal objective [32]. Cross-modal artifact rejection—the original clinical reason for recording both streams—remains almost unexploited computationally [25], and no public *paired* video–EEG animal benchmark exists to test it on [42]. Finally, monitoring that runs for weeks makes streaming, on-device inference a practical prerequisite [31]. App. F develops each of the five in full.

7. Conclusion

We presented a large-scale empirical study of video and EEG deep learning for automated rodent seizure detection and severity grading, built on a pooled twenty-subject synchronized cohort of ~24k clips. Under a single controlled, leakage-aware protocol, and across ten video backbones and seven EEG models, four results stand out: a standard Kinetics-pretrained R(2+1)D backbone *matches* a strong EEG reference for seizure *detection* (0.993 vs. 0.992 AU-ROC) from a non-invasive camera alone; detection is essentially backbone-agnostic, so a 4M model ties a 33M one; video *degrades more gracefully than EEG on severity grading*, the one regime where the two clearly separate; and Grad-CAM confirms the detector attends to genuine behavior. The same backbone reaches 0.979 F1 on the public RodEpiL benchmark once a raw-release encoding artifact is corrected. Together these findings show that the behavioral half of the video–EEG gold standard is far more automatable than its scarce use in animal studies suggests. The natural next step is principled, artifact-aware *fusion* of the two modalities—validated on shared benchmarks and evaluated

across subjects—for which this study provides both the motivation and the single-modality baselines.

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A. Supplementary: A Survey of Video and Video-EEG Seizure Detection

This appendix provides the comprehensive literature survey that motivates the empirical study in the main paper. We organize prior work along two axes, modality (EEG-only, video-only, fused video-EEG) and subject (human vs. animal), and review the video-understanding backbones that make behavioral read-out tractable. The main-paper Related Work (Sec. 2) is a condensed positioning drawn from this material.

A.1. Scope and Taxonomy

This subsection fixes the scope of the survey and of the positioning in Sec. 2. We restrict attention to *automated* seizure detection (methods that consume raw or lightly processed EEG, video, or both, and output a seizure/non-seizure label or a severity grade) and organize it along two axes.

Axis 1: modality. (i) *EEG-only* methods dominate historically and clinically. They range from hand-crafted spectral features fed to classical classifiers to end-to-end recurrent and convolutional networks. (ii) *Video-only* methods read behavior directly through pose estimation or action recognition, and their non-invasiveness makes them attractive. (iii) *Fused video-EEG* methods combine both streams and, in principle, inherit the strengths of each.

Axis 2: subject. Human clinical epilepsy monitoring units generate the largest annotated corpora and host most multimodal work. *Animal* (rodent, canine, porcine) models drive preclinical drug discovery, yet automated multimodal methods there remain scarce.

Table 7 previews where the field is mature and where it is open. The EEG-only cells are crowded. The video-only cells are filling in for humans and now emerging for animals, with a first public rodent benchmark appearing in 2025 [42]. The fused video-EEG animal cell remains nearly empty: the gap that motivates our empirical study.

A.2. EEG-Only Automated Detection

This subsection expands the EEG-only paragraph of Sec. 2. Automated EEG analysis is the incumbent and sets the accuracy bar every video and fusion method is measured against.

Classical machine learning. Early rodent systems paired hand-crafted features with classical classifiers. Bergstrom *et al.* [6] used wavelet decomposition and signal variation to classify spikes and seizures in mouse EEG at 99% accuracy and 91% precision against expert scoring, establishing that automation was viable but tightly coupled to manual feature engineering. Broad reviews of EEG signal processing for epilepsy [13, 59] catalog the dozens of spectral, entropy, and connectivity features this era produced.

Deep learning. The field has since moved to representation learning. Kotloski [27] fed spectrogram images of rat

Table 7. Research gap analysis. Rows are ordered from mature (top) to open (bottom). The fused video-EEG animal setting is barely populated.

Setting		State of the art	Gap
EEG-only, animal		GRU/LSTM, DCNN [18, 27]	No video
Video-only, human		Object/action recog. [23]	Nascent
Video-only, animal		PTZ ML [45]; RodEpil benchmark [42]; this work	Severity grading; EEG fusion
Video+EEG, human		vEpiNet, synchronized [9, 31]	Few, single-site
Video+EEG, animal	2 isolated, binary [35, 40]		Severity; benchmarks

EEG to a pretrained GoogLeNet, reaching 95.6% sensitivity across >2,200 hours in a post-traumatic epilepsy model: an early demonstration that image-based EEG representations and off-the-shelf CNNs transfer to this domain. Ganguly *et al.* [18] applied GRU and LSTM recurrent networks to a chemically diverse rat cohort (Soman/GD and Kainate), showing recurrent models substantially outperform Random Forest and XGBoost, with GRU recall exceeding 90% on most animals. Critically, all of these operate on EEG alone and discard the synchronized video that was recorded alongside it: the reference family we adopt (and broaden to seven models) in the main paper.

A.3. Video-Only Seizure Detection

This subsection expands the video-only paragraph of Sec. 2. Reading seizures from video reframes detection as *action recognition*, a domain computer vision has advanced dramatically. We first summarize the backbones that make this tractable, then their application to seizures.

A.3.1. Video-Understanding Backbones

Modern seizure-video systems inherit one of a few architectural lineages. Two-stream networks fuse an appearance stream with an optical-flow motion stream [48]. Recurrent CNNs (LRCN) attach an LSTM to per-frame CNN features to model temporal order [14]. 3D convolutional networks such as I3D learn spatiotemporal filters directly and, pretrained on Kinetics [24], transfer strongly across video tasks [10]. The R(2+1)D factorization [51] decomposes each 3D kernel into a 2D spatial convolution followed by a 1D temporal convolution, matching 3D accuracy with easier optimization; we adopt it as the default backbone in the main paper. SlowFast [17] processes video at two frame rates to

separate slow appearance from fast motion. Video transformers (TimeSformer [7] and the self-supervised Video-MAE [50]) now lead general benchmarks and reduce the labeled-data appetite that preclinical work cannot easily satisfy. Orthogonally, markerless animal pose estimators, DeepLabCut [38] and SLEAP [41], extract skeletons that make convulsive motion explicit and low-dimensional: an especially attractive front end for rodents.

A.3.2. Human and Animal Seizure Video

Ahmedt-Aristizabal *et al.* [3] analyzed body motion to classify epileptic seizures from clinical video. Karacsony *et al.* [23] combined object and action recognition for the same task. More recently, Xu *et al.* [57] introduced VSViG, a skeleton-based spatiotemporal vision-graph model for real-time video seizure detection at a mainstream vision venue (ECCV 2024). These clinical video systems all target *binary* seizure recognition and stop short of graded severity. The broader deep-learning-for-seizure-video review of Ahmedt-Aristizabal *et al.* [4] argues such methods can reduce the screening burden of long-term video-EEG for both patients and animals. That review also flags the field as still nascent. In animals, the standard behavioral read-out is the Racine scale [43], which grades motor manifestations (twitching, rearing, rhythmic jerking) but is subjective and low-throughput. Sabnis *et al.* [45] took an early substantial supervised-learning step here, predicting ictal events and time-varying severity directly from video in PTZ-induced mice, without implanted electrodes. Their work is promising but (i) uses a stereotyped chemical model, (ii) ignores EEG entirely, and (iii) targets severity rather than robust binary detection: the gaps our study probes on a more complex GD/Kainate cohort. Ren *et al.* [44] similarly built a camera-only convulsive-seizure detector (EPIDetect) for chronically epileptic mice to accelerate drug screening, again framed as binary event detection. Most recently, Perlo *et al.* [42] released *RodEpiL*, the first public benchmark for video-only rodent seizure detection: 13,053 ten-second overhead clips (2,952 seizure, 10,101 normal) from 19 pilocarpine-treated rats. A TimeSformer [7] baseline reaches 97% F1 under strict subject-wise five-fold cross-validation. *RodEpiL* is closely aligned with our own overhead, per-cage-cropped setup and, crucially, evaluates across held-out *subjects*: the honest protocol we adopt for external validation. It remains video-only. Rodent video-EEG *fusion* has been attempted only in isolation, which we turn to next.

A.4. Multimodal Video-EEG Fusion

This subsection expands the fusion paragraph of Sec. 2. Despite video-EEG being the clinical gold standard, remarkably few studies have *computationally* fused both streams: only a handful in the entire human literature (Table 8), and just two isolated attempts in animals. Mullen *et al.* [40] combined video, electrocorticography, and piezoelectric motion

in rats through weighted late fusion of independently trained per-modality models. Concurrently with our work, Lu *et al.* [35] aligned self-supervised EEG and video features with cross-attention in mice (EEGVFusion). Both, however, address only *binary* seizure detection; neither grades severity, the harder three- and five-class problem our study takes on. The broader scarcity traces to the limited availability of paired, annotated video-EEG corpora [60]. The human studies below supply the closest design precedents for a pre-clinical fusion system.

From early fusion to anomaly detection. Aghaei *et al.* [1] is among the earliest to combine video and EEG recordings for automated seizure detection, establishing the core premise that the behavioral and electrographic streams carry complementary evidence. Martini *et al.* [37] reframed the problem as *deep anomaly detection* over paired stereo-EEG and video, learning a model of normal activity that flags seizures as deviations: an unsupervised design that sidesteps the scarcity of seizure labels and is directly relevant to data-poor animal settings.

Synchronized and system-level fusion. Cao *et al.* [9] targeted pediatric epilepsy with synchronized video and EEG streams. Behavioral manifestations in children are subtle and heterogeneous. Their system underscores the temporal-alignment machinery any synchronized setup requires. It also shows the value of adaptively weighting a briefly unreliable modality, such as occluded video or movement-corrupted EEG. The most complete system, vEpiNet [31], detects the patient in-frame, encodes motion via frame differencing and keypoints, encodes EEG spectrograms with EfficientNetV2, and fuses via an MLP. It was trained on $\sim 25k$ IED and $\sim 166k$ non-IED segments from 484 patients, then tested prospectively over 215 hours. It reached AUROC 0.9902 and cut false positives by nearly a third versus its EEG-only counterpart, at 5.7 minutes of compute per hour of recording: the closest methodological reference for a pre-clinical system.

Fusion taxonomy. Reviews organize fusion into early (feature concatenation), mid-level (shared latent space), and late (score-level) strategies [2, 25]. A recurring limitation is that most methods assume equal, fixed modality weights, making them brittle to noisy channels and misalignment [25]: a concern amplified in freely moving animals, where grooming and locomotion generate both video and EEG artifacts.

A.5. The Data Bottleneck

The scarcity described here is what motivates the data-related open problems of Sec. 6. Every survey of this field converges on the same limiting factor: paired, annotated video-EEG data are scarce [2, 60]. Human multimodal corpora sit behind clinical privacy constraints, and the largest reported study still trained on a single institution’s recordings [31]. Animal data sidestep the privacy barrier at the

Table 8. Representative studies that computationally fuse video and EEG for *human* seizure analysis. Animal-model fusion remains nascent (two isolated attempts [35, 40], both limited to binary detection) the gap our severity-grading study is designed to motivate.

Study	Year	Method / vision component	Key result
Aghaei <i>et al.</i> [1]	2017	Combined video and EEG recordings for detection	Early video+EEG fusion demonstration
Martini <i>et al.</i> [37]	2021	Deep anomaly detection, paired stereo-EEG + video	Label-free seizure flagging
Cao <i>et al.</i> [9]	2024	Synchronized video-EEG, pediatric cohort	Detection in childhood epilepsy
Lin <i>et al.</i> (vEpiNet) [31]	2024	Patient detection + frame-diff/keypoints + EfficientNetV2, MLP fusion	AUROC 0.9902; false positives $-1/3$

cost of three practical burdens: expensive implantation, uneven seizure burden across subjects (as in our cohort), and camera geometry that differs per cage and forces per-camera cropping before subjects can be pooled. Two routes past this annotation gap are promising: self-supervised video pretraining [50] and cross-species/cross-modality transfer, such as the multi-space alignment of Canine-EEG-helps-Human [53]. Both remain essentially untried for preclinical video, so the most promising escape from the data bottleneck stays open for the community to pursue.

B. Model Architectures and Training Details

For completeness we describe the video and EEG models compared in Sec. 5 and their shared training recipe. All are standard, off-the-shelf architectures; we change only the classification head and the input adaptation, so the study measures modality and backbone *family* instead of bespoke engineering. Within each modality the models are chosen to span distinct families rather than to tune one, so that a result holding across a whole sweep is a statement about the task and the modality, not about an architecture. Parameter counts for the video backbones are those reported in Table 2.

B.1. Video backbones

The ten video networks span every major action-recognition paradigm. Three axes separate them: how spatial and temporal information are mixed, how the receptive field grows with depth, and how much structural prior the architecture imposes before it sees data. Every backbone is initialized from Kinetics-400 [24] weights and consumes its native clip length and resolution (16–32 frames, 112–224 px); R(2+1)D uses 16 frames at 112×112 . Only the final linear layer is resized to the task’s class count.

Factorized and separable convolution. R(2+1)D-18 [51] (33M), our default, replaces each 3D kernel with a 2D spatial convolution followed by a 1D temporal convolution. The intermediate width is chosen so that the factorized block costs about what the 3D block it replaces would, so the gain comes

from the extra nonlinearity between the spatial and the temporal step and from an easier optimization landscape, not from a smaller budget. S3D [55] (8M) applies the same separation to an Inception-style network and keeps 3D convolution only in the deeper layers, where features are already spatially coarse; this “top-heavy” arrangement is what brings it to roughly a quarter of R(2+1)D’s size. Both impose a strong locality prior: a unit’s receptive field grows by a fixed amount per layer, so long-range temporal structure has to be assembled hierarchically.

Efficient convolution. X3D-M [16] (4M) is grown from a small 2D image classifier by expanding one axis at a time (clip duration, frame rate, spatial resolution, width, depth) and keeping whichever expansion buys the most accuracy per unit of compute. The result is far narrower than its peers, and its role in the sweep is to test whether detection needs the capacity the larger backbones carry.

Two-pathway convolution. SlowFast-R50 [17] (35M) is the one design that treats appearance and motion as separate problems. A Slow pathway samples few frames but carries many channels, modeling what is in the scene; a Fast pathway samples many frames with few channels, modeling how the scene changes; lateral connections inject the fast features into the slow ones. On a task whose label is defined by motion this is the most obviously well-matched prior in the sweep, and SlowFast does take the best detection AUROC (Table 2).

Hierarchical and multiscale attention. Video Swin-T/S [34] (28M/50M) replace convolution with self-attention but keep its habits of locality and hierarchy: attention is computed inside local 3D windows, the windows shift between consecutive blocks so information crosses their boundaries, and resolution drops stage by stage. MViTv1-B/MViTv2-S [15, 30] (36M/34M) reach a similar hierarchy by pooling the attention keys and values, so spatiotemporal resolution falls while channel width rises and a feature pyramid forms inside the transformer; MViTv2 adds decomposed relative position embeddings and residual pooling connections. These four hold a weaker locality prior

than the convolutional models and a much stronger one than global attention.

Global attention and self-supervised pretraining. TimeSformer [7] (121M) drops convolution entirely and uses divided space–time attention (a temporal attention over the same patch across frames, then a spatial attention within each frame) so every unit sees the whole clip from the first block. VideoMAE-B [50] (86M) keeps a plain, non-hierarchical transformer and takes its prior from pretraining instead, having first learned to reconstruct video with over 90% of its spatiotemporal tubes masked. These two carry the least architectural bias and the most parameters, and they finish last and second-to-last at detection (Table 2); we read that as the price of discarding the locality prior at this data scale.

Across the five families the built-in prior ranges from strong locality to none, and capacity spans a factor of thirty, from 4M to 121M. That detection macro-F1 nonetheless stays inside a 0.03 band is what licenses reading the task, rather than the architecture, as the binding constraint (Sec. 5).

Table 9. Video backbone specifications: design paradigm and parameter count for the ten networks compared in Table 2, in the same row order (descending detection macro-F1). Smallest model in **bold**, second-smallest underlined. Capacity spans a factor of thirty, from 4M to 121M, while detection accuracy spans 0.03 macro-F1.

Backbone	Paradigm	Par.
MViTv2-S [30]	multiscale	34M
SlowFast-R50 [17]	two-pathway	35M
R(2+1)D-18 [51]	conv	33M
X3D-M [16]	eff. conv	4M
MViTv1-B [15]	multiscale	36M
Video Swin-S [34]	hier. attn.	50M
S3D [55]	sep. conv	<u>8M</u>
Video Swin-T [34]	hier. attn.	28M
VideoMAE-B [50]	self-sup. ViT	86M
TimeSformer [7]	full attn.	121M

B.2. EEG models

The seven EEG models span the ways a one-dimensional trace can be modeled: recurrently, convolutionally, with attention, or with no learned representation at all. All consume the first recorded channel decimated to 125 Hz and cut into 6 s windows at 3 s stride (750 samples), z-scored per window; a clip decision is the mean of the per-window class posteriors.

Recurrent. The GRU [18] (the architecture of the rodent-EEG detector we adopt as our reference) and the LSTM [20] (both hidden size 128) read the window as a length-750 se-

Table 10. Seed replication of the video backbone comparison: macro-F1 as mean $\pm$ std over five seeds (1, 2, 3, 5, 42), for the ten networks of Table 2 in the same row order. Every cell is $n=5$. Table 2 reports the seed-42 run so that a given model carries one number across Tables 2, 4 and 3; this table gives the noise those point values sit on. The spread across backbones exceeds the median spread within one by $14\times$ at detection (band 0.027 against a median deviation of 0.002), $8\times$ at 3-class (0.065 / 0.009) and $10\times$ at 5-class (0.089 / 0.009), so the ordering is not an artifact of initialization. Two rows repay attention: X3D-M is the strongest model at 3-class and the weakest at 5-class with small deviations at both, a stable property rather than an unlucky run; and VideoMAE-B is the least stable network by a wide margin at 3-class.

Backbone	Detection	3-class	5-class
MViTv2-S [30]	0.972 ± 0.001	0.786 ± 0.005	0.660 ± 0.016
SlowFast-R50 [17]	0.973 ± 0.003	0.789 ± 0.009	0.675 ± 0.008
R(2+1)D-18 [51]	0.971 ± 0.001	0.783 ± 0.002	0.667 ± 0.010
X3D-M [16]	0.969 ± 0.002	0.794 ± 0.005	0.586 ± 0.003
MViTv1-B [15]	0.969 ± 0.002	0.779 ± 0.011	0.653 ± 0.008
Video Swin-S [34]	0.966 ± 0.003	0.771 ± 0.009	0.645 ± 0.011
S3D [55]	0.967 ± 0.002	0.764 ± 0.004	0.636 ± 0.009
Video Swin-T [34]	0.963 ± 0.001	0.766 ± 0.008	0.632 ± 0.002
VideoMAE-B [50]	0.958 ± 0.004	0.729 ± 0.030	0.636 ± 0.017
TimeSformer [7]	0.945 ± 0.001	0.735 ± 0.013	0.601 ± 0.009

quence one sample at a time, carrying a hidden state whose gates decide at each step what to retain and what to overwrite. The LSTM keeps a separate cell state and three gates; the GRU merges these into two, trading a little capacity for fewer parameters. Neither is given any notion of frequency, so whatever spectral structure they exploit has to be inferred from the raw trace, and the recurrence is sequential, so unlike the convolutional and attention models they cannot be parallelized across the window.

Compact convolutional filter bank. EEGNet [29] writes the structure of a classical EEG pipeline into its layers: a first temporal convolution acts as a learned bank of bandpass filters, a depthwise convolution learns a spatial filter for each band, and a separable convolution summarizes the result: all at a deliberately tiny parameter count. The design targets multi-electrode BCI montages, so our single-channel adaptation exercises only its temporal-filtering stage, leaving the depthwise spatial filters with one channel to combine. That is the most plausible reason it is the weakest of the five deep models here (Table 4).

Convolution–attention hybrid. The EEG Conformer [49] splits the problem in two: a convolutional module turns the raw window into a sequence of local temporal tokens, and a self-attention encoder then relates those tokens across the whole window. It is the only model in this set that com-

pares distant positions within a window directly and content-dependently, without a carried state or a growing receptive field.

Dilated convolution. The TCN [5] stacks causal convolutions whose dilation doubles with depth, so the receptive field grows exponentially in the number of layers while the computation stays convolutional and fully parallel. It therefore spans the whole 6 s window within a few blocks and without the sequential bottleneck of a recurrent network, and it is the strongest EEG model on this cohort by macro-F1 (Table 4).

Classical features with tree ensembles. Random Forest [8] (100 trees, balanced class weights) and XGBoost [12] (300 trees, depth 6, learning rate 0.1) learn no representation at all. They are trained not on the raw window but on a fixed eleven-number per-window descriptor: five relative band powers (delta 0.5–4, theta 4–8, alpha 8–13, beta 13–30, gamma 30–60 Hz) plus six time-domain statistics (mean, standard deviation, RMS, line length, activity, zero crossings). These are the quantities hand-built preclinical seizure detectors have used for decades. Random Forest averages independently grown trees; XGBoost grows them in sequence, each fitting the residual of the ensemble so far. Their role is to test whether the deep models earn their complexity, and Sec. 5 finds that they only partly do. That a fixed descriptor stays competitive with learned representations suggests the ceiling here is set by what one decimated channel carries rather than by model capacity, which is why we read the single-channel montage as a bound on the reported cross-modal margin and not as a weakness of the EEG models themselves.

B.3. Training and evaluation protocol

All models share the session-disjoint split (seed 49), inverse-frequency-weighted objective, macro-averaged metrics, and (for EEG) the window→clip mean-pool of Sec. 3. Video backbones are fine-tuned with AdamW (learning rate 3×10^{-4} , weight decay 10^{-4}), cosine annealing, mixed-precision, batch size 16, for a matched 12-epoch budget; the two attention models that were unstable at this budget (Video Swin, MViTv2 on RodEpiI) used a reduced learning rate of 10^{-4} . Deep EEG models are trained with AdamW (learning rate 10^{-3} , weight decay 10^{-4}), cosine annealing, batch size 256, for 30 epochs. The deep models optimize inverse-frequency-weighted cross-entropy; the classical models use balanced class weighting. All video and deep-EEG runs share the split seed across modalities so the head-to-head numbers in Sec. 5 are directly comparable.

B.4. Metric definitions

Every score in the paper is computed from the validation confusion matrix of a single run, then aggregated over runs where indicated. We give the definitions here so that the

macro-F1 vs. MCC contrast drawn in Sec. 3 and Sec. 5.2 is unambiguous.

Notation. For a K -class task let $C \in \mathbb{N}^{K \times K}$ be the confusion matrix, where C_{ij} counts validation clips of true class i predicted as class j , and let $N = \sum_{i,j} C_{ij}$. Write $t_k = \sum_j C_{kj}$ and $p_k = \sum_i C_{ik}$ for the number of clips whose true and predicted class is k , and

$$\begin{aligned} \text{TP}_k &= C_{kk}, & \text{FP}_k &= p_k - C_{kk}, \\ \text{FN}_k &= t_k - C_{kk}, & \text{TN}_k &= N - p_k - t_k + C_{kk}. \end{aligned} \quad (2)$$

Precision, recall and F1. Per class,

$$P_k = \frac{\text{TP}_k}{\text{TP}_k + \text{FP}_k}, \quad R_k = \frac{\text{TP}_k}{\text{TP}_k + \text{FN}_k}, \quad (3)$$

$$F_{1,k} = \frac{2P_k R_k}{P_k + R_k} = \frac{2 \text{TP}_k}{2 \text{TP}_k + \text{FP}_k + \text{FN}_k}. \quad (4)$$

The columns labelled P, R and F1 are the unweighted means over classes,

$$P_{\text{macro}} = \frac{1}{K} \sum_{k=1}^K P_k, \quad \text{macro-F1} = \frac{1}{K} \sum_{k=1}^K F_{1,k}, \quad (5)$$

and likewise for recall. Macro-F1 is the mean of the per-class $F_{1,k}$ rather than the F_1 of P_{macro} and R_{macro} , which differ once the classes are unbalanced; macro recall coincides with balanced accuracy. On RodEpiI (Tables 6 and 15) we also report the seizure-class score, Eq. (4) evaluated at the positive class alone, to match the convention of the original release [42].

AUROC. For a binary task with score s_i assigned to the positive class, and $\mathcal{P}, \mathcal{N}$ the positive and negative clip sets, AUROC is the probability that a random positive outranks a random negative,

$$\text{AUROC} = \frac{1}{|\mathcal{P}||\mathcal{N}|} \sum_{i \in \mathcal{P}} \sum_{j \in \mathcal{N}} \left[\mathbf{1}(s_i > s_j) + \frac{1}{2} \mathbf{1}(s_i = s_j) \right]. \quad (6)$$

At grading we report the macro one-vs-rest average: Eq. (6) is applied with class k as the positive set and the rest as the negative set, then averaged over k . AUROC is threshold-free and reads the posteriors directly, whereas the other metrics act on the arg-max decision.

Matthews correlation coefficient. MCC [39] is the Pearson correlation between the true and predicted label indicators. In its K -class form [19],

$$\text{MCC} = \frac{N \sum_k C_{kk} - \sum_k p_k t_k}{\sqrt{(N^2 - \sum_k p_k^2)(N^2 - \sum_k t_k^2)}}, \quad (7)$$

which at $K=2$ reduces to the familiar

$$\text{MCC} = \frac{\text{TP} \cdot \text{TN} - \text{FP} \cdot \text{FN}}{\sqrt{p_1 p_2 t_1 t_2}}. \quad (8)$$

MCC lies in $[-1, 1]$ and is 0 at chance, where Eqs. (3)–(5) lie in $[0, 1]$. Equation (7) takes no per-class average: every clip enters through the sums, so frequent classes carry proportionally more weight. This is the formal sense in which the two summaries differ (Sec. 5.2): a rare stage with poor $F_{1,k}$ moves macro-F1 by $1/K$ of its deficit however few clips it holds, but moves Eq. (7) only in proportion to t_k .

Aggregation over runs. Where a table reports mean $\pm$ std over M runs, whether five seeds (Table 1) or five cross-validation folds (Tables 5 and 15), each metric is computed per run from that run’s own confusion matrix and then summarized as

$$\bar{x} = \frac{1}{M} \sum_{m=1}^M x_m, \quad \sigma = \sqrt{\frac{1}{M-1} \sum_{m=1}^M (x_m - \bar{x})^2}, \quad (9)$$

with $M = 5$ throughout. Folds are never pooled into one confusion matrix, so $\bar{x}$ is a mean of per-fold scores rather than a score over concatenated predictions.

Macro-F1 vs. MCC on our results. Sec. 5.2 states that the two summaries disagree about the difficulty of grading and agree about every ordering; here is that comparison in full. At 5-class the video model scores 0.662 macro-F1 against 0.830 MCC, which places most of the macro-F1 decline in Stage 4–5 rather than in a general breakdown (Table 17): by Eq. (5) the two rare stages contribute two fifths of macro-F1 while holding 201 of 5,289 clips, and Eq. (7) weights them by that count instead. The orderings nonetheless coincide. The cross-modal gap widens with granularity under both ($0.004 \rightarrow 0.041 \rightarrow 0.128$ macro-F1; $0.008 \rightarrow 0.063 \rightarrow 0.128$ MCC, Table 1), the ten video backbones stay within a narrow band under both (Table 2), and video remains the more predictable modality across held-out animals, with a per-fold MCC spread of ± 0.031 against the EEG reference’s ± 0.101 (Table 5). No conclusion in the paper turns on which of the two is used.

B.5. EEG window, stride, rate and pooling

The video side has a temporal ablation (Table 3); this is its EEG counterpart, and it exists to check that the EEG reference of Sec. 5.2 is limited by the signal rather than by our preprocessing choices. Holding the GRU, the cohort and the session-disjoint split fixed, we sweep four axes at 5-class grading (Table 11): window length, window stride, sampling rate, and the rule that turns per-window posteriors into a clip decision. The three temporal axes are staged rather than crossed (stride and rate are swept at the best window) because each cell needs its own window cache. Pooling is free by comparison: it applies after training, so all five rules are scored from the same saved window posteriors.

The temporal parameters barely move macro-F1. Across windows from 2 to 8 s, macro-F1 under mean pooling moves only between 0.337 and 0.352, with 4 s marginally ahead of

the 6 s default. Halving or doubling the 125 Hz rate changes almost nothing (0.341 on either side of 0.352). Denser striding helps a little (a 1 s stride reaches 0.367 against 0.331 at 4 s) which is consistent with more windows per clip giving the pooling stage more to average. MCC disagrees on the window axis: it spans 0.341–0.392 there and puts the 6 s default ahead of 4 s. Macro-F1 weights all five stages equally while MCC follows the frequent ones, so the flatness above is a property of the rare-stage-weighted view rather than of window length itself.

Under macro-F1 the pooling rule matters most. Log-mean pooling beats plain mean at all eight configurations, by 0.022–0.049 macro-F1, and the best cell in the sweep (4 s window, 2 s stride, 125 Hz, log-mean) reaches 0.402, against 0.340 for the 6 s / mean configuration the main paper uses, measured in this same sweep. Max pooling is far worse everywhere (0.197–0.234): a single high-scoring window is a poor summary of a clip in which most windows are uninformative. Retraining the GRU from scratch at the best window and stride confirms the sweep without improving on it: 4 s windows at 1 s stride reach 0.367 ± 0.017 macro-F1 over five seeds, which matches the corresponding swept cell (0.367) and leaves the 0.402 figure attributable to the pooling rule alone.

None of this changes the cross-modal picture. The best EEG configuration we found still trails video by 0.26 macro-F1 at 5-class (0.402 vs. 0.662). We keep the shared 6 s / 3 s / 125 Hz / mean setting in the main paper because all seven EEG models are trained under it, and re-tuning one of them alone would break the comparison in Table 4.

C. Extended Dataset Details

This appendix provides the full per-subject composition of the pooled twenty-subject cohort together with the visual examples referenced from the Dataset and Experiments sections (Secs. 4 and 5).

C.1. Per-subject cohort composition

Table 12 first gives the corpus in recorded time, a dimension the main paper reports only in clips. Clip length is not fixed: each clip spans its annotated event plus a 10 s buffer at either end, so duration tracks seizure duration and averages 77 s. The two classes are near-identical in both count and total time, which is what the matched-negative sampling of Sec. 4 is designed to produce.

Table 13 lists, for every animal, the number of recording sessions, total clips, the non-seizure vs. seizure split, and the Racine stage distribution of the seizure clips; Figure 9 plots the same counts as stacked bars. Two properties of the cohort motivate our evaluation protocol (Sec. 3). First, seizure burden is *severely long-tailed*: the four busiest animals (R1–R4) each contribute more than 2,700 clips and together supply nearly half of the corpus, whereas the

Table 11. EEG preprocessing sweep (GRU, 5-class grading, macro-F1, identical cohort and session-disjoint split). Columns are the rule used to pool per-window posteriors into a clip decision; rows sweep window length, stride and sampling rate. Stride and rate are swept at the best window (4 s), so the marked row is one run shared by all three panels. The temporal axes move macro-F1 by at most 0.036, whereas switching mean for log-mean pooling gains up to 0.049, so the best cell in the sweep reaches 0.402. The final column gives MCC under mean pooling, the one rule for which per-clip decisions were stored; it reorders the window panel, placing the 6 s default ahead of 4 s. Best per column in **bold**.

	Macro-F1 by pooling rule					MCC (mean)
	mean	max	top- <i>k</i>	log-mean	attn	
<i>Window length (stride = win/2, 125 Hz)</i>						
2 s	0.351	0.227	0.266	0.374	0.350	0.369
4 s [†]	0.352	0.205	0.272	0.402	0.339	0.367
6 s (default)	0.340	0.220	0.289	0.388	0.340	0.392
8 s	0.337	0.197	0.268	0.377	0.333	0.341
<i>Stride (4 s window, 125 Hz)</i>						
1 s	0.367	0.234	0.288	0.398	0.360	0.397
2 s [†]	0.352	0.205	0.272	0.402	0.339	0.367
4 s	0.331	0.218	0.263	0.371	0.315	0.363
<i>Sampling rate (4 s window, 2 s stride)</i>						
62.5 Hz	0.341	0.220	0.278	0.363	0.333	0.347
125 Hz [†]	0.352	0.205	0.272	0.402	0.339	0.367
250 Hz	0.341	0.210	0.280	0.362	0.352	0.360

[†] the same run, shared by the three panels.

Table 12. Corpus scale in recorded time, measured from the per-clip timing records. Each clip spans its annotated event plus a 10 s pre- and post-buffer, so mean duration reflects mean event duration. Seizure and matched non-seizure clips are balanced in count and in total time. Measured over 20 subjects and 601 recording sessions; the trainer loads 24,497 of these clips and splits them 19,208/5,289 session-disjoint (Sec. 3).

Clip class	Clips	Hours	Mean (s)
Seizure	12,228	261.4	77.0
Matched non-seizure	12,344	263.7	76.9
Total	24,572	525.1	77.0

quietest (R18–R20) contribute only 11–29 clips apiece: a span of nearly three orders of magnitude. A single pooled score is therefore dominated by a handful of animals, which is why we report per-subject breakdowns throughout and adopt subject-disjoint cross-validation for the generalization study (Sec. 5). Second, the *severity* distribution is itself long-tailed: Stage 3 accounts for the majority of seizure clips (9,256) while the most severe Stage 5 is represented by only 193 clips across the entire cohort, concentrated in a few animals. This scarcity of severe events is the central diffi-

culty of fine-grained grading and explains why the rare-stage scores in Sec. 5 are the weakest regime for both modalities.

C.2. Task and per-stage visual examples

Figure 10 makes the binary detection task concrete. The top row is drawn from a non-seizure interval and the bottom row from a seizure of the *same* rat in the *same* recording session, so camera, cage and lighting are identical across the pair and any separation the detector achieves has to come from behavior. This is the matched-negative sampling our clip extraction enforces throughout (Sec. 4), and it is what makes the detection scores of Sec. 5 a test of behavior rather than of scene appearance.

Figure 11 shows per-stage seizure *filmstrips*: five frames sampled across one representative clip for each Racine stage (rows S2–S5), time increasing left to right. The behavioral signal becomes progressively more overt with stage, from the subtle movements of Stage 2, through rearing at Stage 3–4, to the pronounced convulsive motion of Stage 5. This visual progression is the cue a video model reads directly and that a single EEG channel cannot see, and it underlies video’s graceful degradation on the grading task (Sec. 5). All frames are from de-identified clips, cropped to remove cage-side identifiers.

C.3. Cross-subject appearance variation

Figure 12 shows one frame from each of six animals. Cage placement, camera framing, illumination (daylight vs. infrared) and coat all differ from animal to animal, because each rat is filmed in its own home cage by its own fixed camera. This is the appearance shift a model meets when it is tested on an animal it has never seen, and it is why held-out-*subject* cross-validation is a harder protocol than held-out-session (Sec. 5). It is also why clips are cropped per cage before subjects are pooled: without that step a model could separate animals on framing alone.

D. Extended Qualitative Results

This appendix collects the qualitative evidence behind two claims of Sec. 5: where the video detector fails, and what it attends to when it succeeds.

D.1. Failure cases

Figure 13 shows the two regimes that account for most of the residual error. The first is the subtle end of the severity scale: a Stage 2 clip whose only motion is a brief twitch is hard to separate from ordinary cage behavior. The second is data scarcity: the near-silent animals at the bottom of Table 13 contribute only a handful of validation clips each, often under low-light infrared capture, so their per-subject scores rest on very little evidence. Both regimes are properties of the cohort rather than of the backbone, which is why we pair every pooled score with a per-subject breakdown (Sec. 3).

Table 13. Per-subject composition of the pooled video–EEG cohort (20 rats). For each animal: recording sessions, total clips, non-seizure vs. seizure clips, and the Racine stage distribution of the seizure clips (S2–S5). Seizure burden is severely uneven: the four busiest animals (R1, R2, R3, R4) supply nearly half of all clips, subject totals span 11–3,091 clips, and the most severe stage (S5, 193 clips) is rare throughout. Counts are the paired clips used for training and evaluation. Subjects are de-identified and ordered by clip count (R1–R20).

Subject	Sess.	Clips	Non-sz.	Seiz.	S2	S3	S4	S5
R1	42	3,091	1,576	1,515	22	1,325	131	37
R2	44	2,830	1,431	1,399	113	1,094	181	11
R3	38	2,774	1,380	1,394	364	873	124	33
R4	42	2,701	1,351	1,350	22	1,062	226	40
R5	43	2,195	1,100	1,095	23	946	114	12
R6	43	1,963	955	1,008	218	782	8	0
R7	32	1,600	814	786	408	317	36	25
R8	43	1,502	754	748	33	648	61	6
R9	45	1,474	763	711	177	511	21	2
R10	39	1,199	631	568	16	544	6	2
R11	24	918	491	427	6	364	47	10
R12	39	841	422	419	10	387	20	2
R13	23	444	224	220	2	174	42	2
R14	25	262	131	131	12	112	3	4
R15	27	118	59	59	3	39	16	1
R16	18	103	53	50	4	36	7	3
R17	14	62	32	30	1	25	2	2
R18	11	29	17	12	0	9	2	1
R19	4	14	7	7	0	4	3	0
R20	5	11	6	5	1	4	0	0
Total	601	24,131	12,197	11,934	1,435	9,256	1,050	193

D.2. Grad-CAM interpretability

This subsection extends the Grad-CAM [47] evidence of Sec. 5.1, where Fig. 5 shows seizure-class activation concentrating on the animal’s body throughout a Stage 4 convulsion while staying diffuse on a non-seizure clip. Because negatives are sampled from the same sessions as positives (Sec. 4), a detector keying on cage background or recording-setup cues would produce similar maps for the two, so the contrast is evidence that the model reads behavior. Figure 15 repeats the check across Stages 2, 3 and 5 and finds the same pattern, so the result is not an artifact of one clip or one severity level.

D.3. What the shortcut evidence does and does not establish

Sec. 5.1 sets out the controls that bound the risk of the detector reading a shortcut rather than behavior. This subsection states their limits.

The structural arguments exclude particular shortcuts (fixed scene content, cage framing, static appearance) yet they cannot exclude cues we did not think to control, such as bedding displacement or a subject-specific motion signature that correlates with seizure burden. Grad-CAM is weaker evidence still: it shows where gradient magnitude

concentrates, which is suggestive of what the model attends to, short of a causal account of what it uses. The strongest indirect check available to us is the subject-disjoint protocol of Sec. 5.2, where a detector leaning on animal identity should degrade sharply on unseen animals and instead gives up only 0.022 macro-F1 at detection. Settling the question would take interventions we have not run (masking the animal, shuffling frame order, or training on background crops) and we regard those as the natural next test, and do not treat the matter as closed.

E. Extended Quantitative Results

This appendix holds two result sets summarized in the main paper: how detection accuracy scales with the training budget and with pretraining (Sec. 5.1), and the full ten-backbone sweep on the external RodEpil benchmark (Sec. 5.2).

E.1. Data efficiency and pretraining

How much of this performance is bought by data and by transfer? Two ablations (Table 14) answer this. First, holding the R(2+1)D backbone fixed and training on a growing fraction of the subjects, detection *saturates quickly*: just two animals (62 clips) already give 0.79 macro-F1, half the cohort (ten subjects, ~11k clips) reaches 0.957, and the full

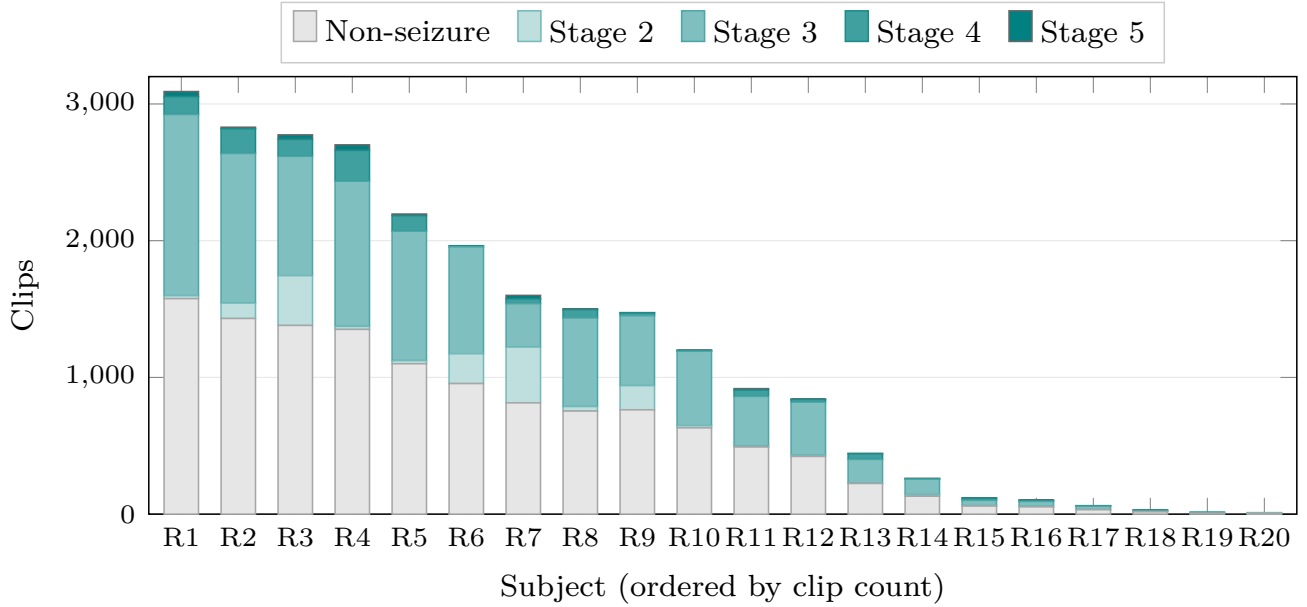


Figure 9. Per-subject clip composition of the pooled cohort (stacked counts from Table 13). Seizure burden is severely long-tailed: the four busiest animals (R1–R4) dwarf the rest, and subject totals span nearly three orders of magnitude (11–3,091 clips). The most severe stages (S4, dark; S5, darkest) form only a thin sliver on every bar (193 Stage-5 clips in the entire cohort) which is why fine-grained grading is so hard.



Figure 10. The binary detection task, visualized. Four frames sampled across a non-seizure interval (top) and across a seizure (bottom) of the *same* rat in the *same* recording session, so camera, cage and lighting are identical across the pair and the detector must separate the two from behavior alone. De-identified clips.

set only adds a further 0.013 to 0.970. The curve is not perfectly monotone (which subjects enter training matters more than their count, so twelve subjects dip slightly to 0.946 before recovering) but the overall trend is a shallow saturation, indicating the task is not data-starved at this scale. Second, pretraining helps but is not essential: training the same backbone from random initialization, with no Kinetics weights, still reaches 0.952 macro-F1, a modest 0.018 below the pre-trained 0.970. Kinetics transfer thus provides a small, consistent boost and is not a prerequisite.

E.2. All ten backbones on RodEpiL

Beyond R(2+1)D we run the full suite on re-encoded RodEpiL under the same subject-wise five-fold protocol and

matched budget (Table 15). Two things stand out. First, the ranking *reshuffles* relative to our cohort and spreads far wider (0.87–0.99 macro-F1 vs. a 0.03 band on our data). R(2+1)D now leads clearly (0.986) and the multiscale MViTv2 stays competitive (0.946, third), but the remaining transformers, TimeSformer (0.871), MViT-v1 (0.890) and Video Swin (0.90–0.91), fall behind the convolutional S3D (0.944) and X3D (0.933). This ordering inverts our cohort’s, where MViTv2 topped (Table 2), reinforcing that backbone choice is a second-order, data-regime-dependent effect rather than a universal ranking. Second, several video backbones *exceed* the dataset’s reported TimeSformer baseline of 0.97 seizure-F1: our R(2+1)D reaches 0.978 here under the matched 12-epoch budget, and 0.979 in the dedi-



Figure 11. Per-stage seizure *filmstrips*: five frames sampled across one representative seizure clip for each Racine stage (rows S2–S5), time increasing left to right. The behavioral signal grows more overt with stage (from mild movement at Stage 2 to pronounced rearing and convulsive motion at Stage 4–5) which is what a video model reads directly and an EEG channel cannot see. Frames are from de-identified clips, cropped to remove cage-side identifiers.

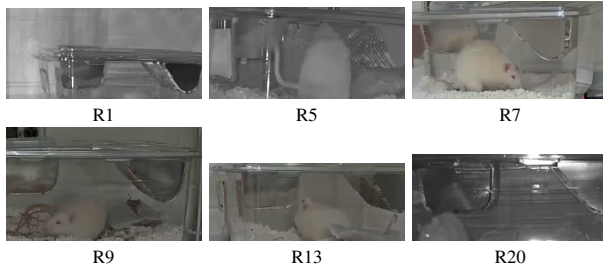


Figure 12. Cross-subject variation: one frame from six animals (de-identified, R = subject). Cage placement, camera framing, lighting (daylight vs. infrared), and coat differ per animal, part of why held-out-*subject* generalization is harder than held-out-session, and why the video detector’s stability across subjects (Fig. 6) matters.

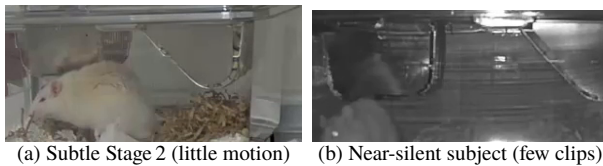


Figure 13. Where video is hardest. (a) A Stage 2 clip with only subtle movement, easily confused with normal behavior. (b) A near-silent subject (bottom rows of Table 13, App. C) with few validation clips and low-light infrared capture. Rare mild events and data-poor animals drive most of the residual error. De-identified clips.

Table 14. Data efficiency and pretraining (R(2+1)D, 2-class detection, session-disjoint). Top: macro-F1 vs. the fraction of training subjects used (validation set fixed). Bottom: Kinetics-pretrained vs. from-scratch. Detection saturates well before the full cohort, and pretraining adds only ~ 0.02 . MCC tracks macro-F1 throughout but falls off faster in the low-data regime.

Train subj.	Frac.	Clips	Macro-F1	MCC
2	0.10	62	0.790	0.583
5	0.25	1,905	0.893	0.787
10	0.50	10,969	0.957	0.914
12	0.60	11,832	0.946	0.892
15	0.75	16,358	0.960	0.921
20	1.00	19,208	0.970	0.941
<i>Pretraining</i> (full cohort):				
Kinetics-pretrained			0.970	0.941
from scratch (random init)			0.952	0.903

the original release under the authors’ own training setup, whereas our numbers use re-encoded clips and a single fixed budget, at which our own TimeSformer reaches only 0.799 seizure-F1. Two caveats reinforce this: SlowFast saturates the benchmark (a perfect 1.000, which we flag rather than celebrate), and the Video Swin and MViTv2 transformers needed a lower learning rate (10^{-4}) than the conv nets to train stably at this budget.

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1683 cated run of Table 6. This suggests the benchmark is near-
1684 solved for strong motion models. This comparison is indica-
1685 tive rather than strict: the 0.97 reference was reported on

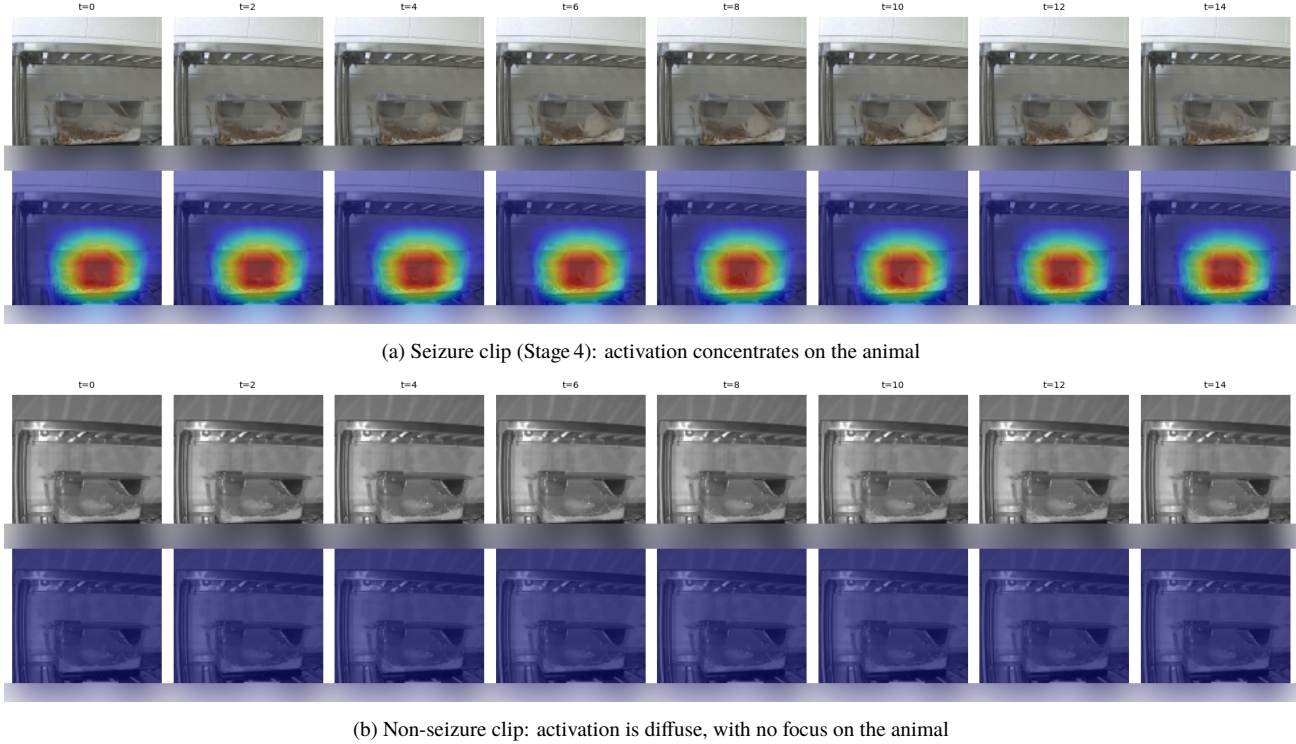


Figure 14. Full-width version of Fig. 5: all eight sampled frames for the same two clips, with input frames (top) and seizure-class activation (bottom) in each strip. De-identified clips; identifier bands redacted.

Table 15. All ten video backbones on re-encoded RodEpiL [42] (subject-wise 5-fold; F1/AUROC are means over folds, MCC is mean $\pm$ std). Convolutional backbones lead and attention models trail: the reverse of our cohort ranking (Table 2); paradigm labels are given in App. B.1, Table 9. [†]SlowFast saturates the benchmark (flagged, not a genuine separation). Best non-saturated result in **bold**; the reference is the dataset’s reported TimeSformer (0.97 seizure-F1).

Backbone	Seiz-F1	Macro-F1	AUROC	MCC
SlowFast-R50 [†] [17]	1.000	1.000	1.000	1.000 $\pm$ 0.000
R(2+1)D-18 [51]	0.978	0.986	0.999	0.972 $\pm$ 0.015
MViTv2-S [30]	0.916	0.946	0.989	0.891 $\pm$ 0.020
S3D [55]	0.912	0.944	0.991	0.889 $\pm$ 0.029
X3D-M [16]	0.895	0.933	0.983	0.866 $\pm$ 0.010
VideoMAE-B [50]	0.874	0.919	0.982	0.839 $\pm$ 0.025
Video Swin-T [34]	0.861	0.910	0.979	0.823 $\pm$ 0.032
Video Swin-S [34]	0.854	0.904	0.977	0.810 $\pm$ 0.031
MViTv1-B [15]	0.835	0.890	0.971	0.784 $\pm$ 0.049
TimeSformer [7]	0.799	0.871	0.957	0.743 $\pm$ 0.038

E.3. Backbone, temporal and spatial ablations in detail

This subsection expands the summary read-outs of Sec. 5.1.

Backbone ranking. Within the 0.03 macro-F1 band of Table 2, the multiscale MViTv2-S and the motion-oriented SlowFast top the table (0.974 macro-F1; SlowFast attains the best AUROC, 0.995), marginally ahead of R(2+1)D and the tiny X3D-M, with TimeSformer trailing the field. The ranking is dataset-dependent: on RodEpiL, R(2+1)D and TimeSformer are at near-parity (our 0.979 vs. the reported 0.97), reinforcing that once the data regime is fixed, architecture is a second-order factor relative to modality and preprocessing. The clustering extends to grading, where the 3- and 5-class columns again fall in narrow bands (3-class macro-F1 0.74–0.79, 5-class 0.59–0.67).

Temporal sampling. Four frames already recover most of the detection signal (0.940) and eight reach 0.960, after which the curve flattens: 16 frames give 0.970 and 32 give 0.972. Grading follows the same shape one level down: three- and five-class macro-F1 both climb from 8 to 32 frames (0.763→0.793 and 0.627→0.667). Doubling to 64 returns nothing at detection or 3-class and actively costs at 5-class, where macro-F1 falls to 0.632, a drop spread across all five classes and largest on Stage 2. Against 32 frames the 16-frame default concedes 0.002 macro-F1 at detection, 0.012 at 3-class and 0.005 at 5-class for half the compute; the detection and 5-class margins fall inside the seed spread of Table 1, so we treat the two budgets as equivalent.

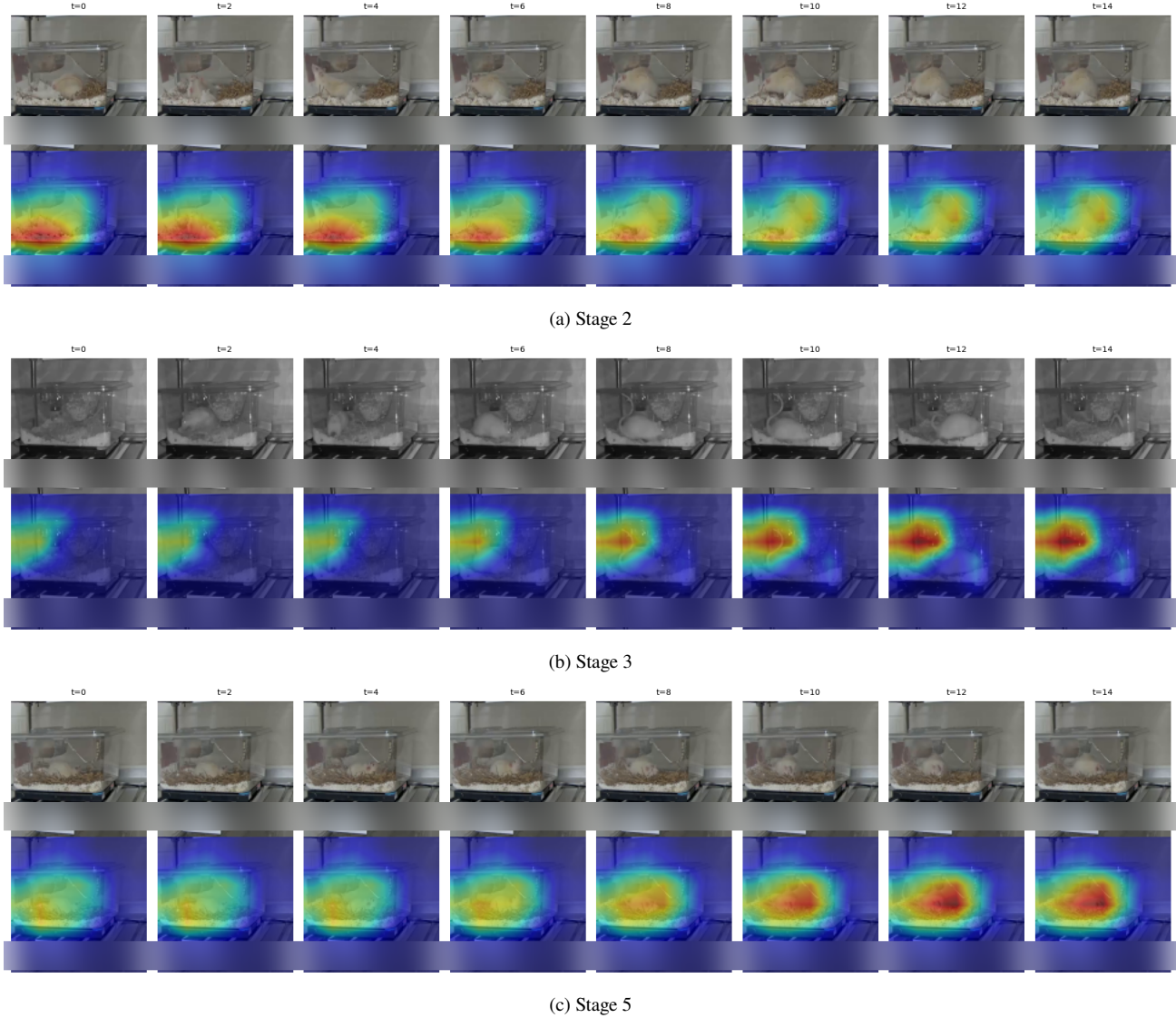


Figure 15. Grad-CAM across severity stages for the R(2+1)D detector; each strip shows input frames (top) and seizure-class activation (bottom). Activation concentrates on the animal consistently across Stages 2, 3 and 5, indicating the interpretability finding is not specific to one clip or stage. De-identified clips (identifier bands redacted).

Spatial resolution. Resolution beyond the 112 default does not help and slightly hurts. Our clips are only ~ 300 – 480 px native and the seizure signal is coarse whole-body motion, so the larger settings upsample the recording and spend compute without adding usable detail. Higher resolutions (512/1024) would upsample further beyond the source and are not meaningful here.

E.4. Where each modality fails, and why

Table 17 shows that the two modalities lose accuracy on different classes. This subsection asks what the errors look like and what produces them. Figure 16 gives the full confusion structure at all three granularities and Figure 17 sum-

marises the analysis; every number below is reproduced by `analyze_failures.py` and `motion_vs_error.py`.

The two failure modes are different in kind. Video’s errors are overwhelmingly *off-by-one* on the severity scale: 72.4% of them land on an adjacent stage, and the dominant single cell is Stage 4 $\rightarrow$ Stage 3, which absorbs 95 of 167 Stage 4 clips (57%). Video reliably registers that convulsive motion is occurring and misjudges its degree. EEG fails in two further ways video does not. First it *loses the event*: 208 of 2064 Stage 3 clips (10.1%) are classified non-seizure, against 49 of 2148 (2.3%) for video: a four-fold higher miss rate on the commonest seizure class. Second, when it does fire, its errors do not respect the ordinal scale: 94 Stage 3

Table 16. Spatial-resolution ablation: R(2+1)D detection at three input sizes (16 frames, session-disjoint split, seed 42, 10 epochs; batch scaled 16/8/4). Source clips are only $\sim 300\text{--}480$ px, so both larger settings upsample the recording and accuracy falls monotonically on all three metrics, at AUROC the top two separate only past the third decimal (0.9934 vs. 0.9926). The 448 run was stopped by an out-of-memory kill after 8 of its 10 epochs; we report its best checkpoint, so that row is if anything an optimistic bound on the larger setting. Best per column in **bold**.

Input size	Macro-F1	AUROC	MCC
112×112 (default)	0.970	0.993	0.941
224×224	0.967	0.993	0.934
448×448	0.952	0.988	0.904

Table 17. Per-class breakdown at 5-class grading, video (R(2+1)D) vs. the EEG GRU (n is the held-out clip count). Both are strong on the common classes and weak on the rare high-severity stages, and the failure is sharper for EEG: Stage 4–5 F1 of 0.20 / 0.12 against video’s 0.36 / 0.42. Over all five classes the run-level MCC is 0.830 for video against 0.702 for EEG. The counts differ slightly between the two columns because the EEG models were retrained on a re-generated clip corpus, so the held-out sets are not clip-identical.

Stage	Video				EEG			
	n	P	R	F1	n	P	R	F1
non-seizure	2720	0.965	0.968	0.966	2737	0.914	0.955	0.934
Stage 2	220	0.681	0.650	0.665	256	0.491	0.863	0.626
Stage 3	2148	0.904	0.896	0.900	2064	0.867	0.717	0.785
Stage 4	167	0.330	0.401	0.362	229	0.268	0.162	0.202
Stage 5	34	0.579	0.324	0.415	33	0.074	0.364	0.123

clips are predicted Stage 5 and 42 Stage 4 clips are predicted Stage 5, so errors of two stages or more account for 44% of EEG’s mistakes against 28% of video’s (mean $|\Delta\text{stage}|$ 1.51 vs. 1.30; Fig. 17a). A single channel supports a seizure/no-seizure decision but does not carry a monotone severity axis, so its mistakes are not near-misses.

The one class EEG wins is the one with almost no behavior. At Stage 2 the ordering reverses: EEG recalls 0.863 against video’s 0.650 (Fig. 17b). Racine Stage 2 is head nodding and facial clonus: the rung of the scale with the least whole-body movement. Restricted to the 2830 clips both models scored (Table 18), EEG is the only correct model on 44 Stage 2 clips while video is the only correct model on 7; at Stage 3 the same comparison runs 286 to 43 the other way. Video’s advantage is behavioral, and it recedes exactly where the behavior does.

EEG needs a strong, sustained event. Two independent measurements point the same way. The seizure a clip is matched to is shorter when EEG is wrong than when it is right (median 40.4 s vs. 45.3 s), while video shows no such

Table 18. Paired outcomes on the 2830 clips present in both models’ held-out sets (5-class; the two sets are not clip-identical, see Table 17). The per-class rows count clips where exactly one modality is correct. Stage 2 is the only class where EEG-only-correct outnumbered video-only-correct.

True class	n	video only	EEG only
non-seizure	1488	73	44
Stage 2	192	7	44
Stage 3	1062	286	43
Stage 4	74	27	8
Stage 5	14	8	1

overall: both correct 2161 (76.4%), video only 401 (14.2%), EEG only 140 (4.9%), both wrong 128 (4.5%)

Table 19. Recall on the low- and high-motion half of each class, split at that class’s median motion energy (900 clips sampled from the shared set). Motion energy is the mean absolute frame difference over 24 evenly spaced frames at 96×96 grayscale, computed over the whole clip including its 10 s pre- and post-buffers, so it is a coarse proxy for how vigorous the seizure was.

True class	n	Video recall			EEG recall	
		med.	low	high	low	high
non-seizure	466	1.53	0.983	0.966	0.906	1.000
Stage 2	58	1.57	0.655	0.586	0.966	0.862
Stage 3	350	1.78	0.891	0.943	0.537	0.777
Stage 4	21	2.17	0.182	0.700	0.182	0.000

dependence (45.1 s vs. 44.2 s). And splitting each class at its median clip motion energy (mean absolute frame difference over a sparse grayscale sample, Table 19) EEG’s Stage 3 recall rises from 0.537 to 0.777 across the split. Since the EEG model never sees the video, motion energy here is standing in for *event intensity*: vigorous seizures produce both large movement and a clearer electrographic and autonomic signature. The same split moves video’s Stage 4 recall from 0.182 to 0.700, which is the mechanism behind the Stage 4 $\rightarrow$ Stage 3 compression above, though that cell rests on only 21 clips.

Two explanations we tested and could not support. We report both, because each is the first hypothesis a reader is likely to form. (i) *Video loses Stage 2 because the motion is subtle.* This is the natural reading of the reversal above, but it does not survive measurement: Stage 2 video recall is 0.655 on the low-motion half and 0.586 on the high, i.e. flat and weakly in the wrong direction (Table 19), and video’s errors overall sit on *higher*-motion clips than its successes (median 1.676 vs. 1.593). Whatever separates the Stage 2 clips EEG

recovers from those video recovers, our motion proxy does not capture it. (ii) *EEG grades poorly because the channel is mostly ECG*. Nineteen of twenty-one animals are recorded as ECG rather than EEG (App. C.1), which would explain a signal that detects seizures but does not encode motor severity. The one true-EEG animal in the held-out set, RN219, does not stand out: detection macro-F1 0.981 and 5-class 0.443, against 0.963 and 0.474 averaged over the fourteen ECG animals, mid-pack at detection and slightly below average at grading. With $n=1$ this cannot refute the hypothesis, but it does not support it either, and we therefore do not rest any claim on it.

F. Limitations and Future Work in Detail

Sec. 6 states five limitations in brief. We develop each of them here, with the evidence from our own results that motivates it.

(1) Cross-subject generalization. Seizure burden and per-animal appearance both vary sharply across the cohort (App. C.1 and App. C.3), which makes held-out-subject evaluation the honest test, beyond held-out-session. We report subject-disjoint five-fold cross-validation on the pooled cohort at all three granularities (Table 5) and subject-wise folds on RodEpi [42] (App. E.2). Both modalities hold up at detection: video moves from 0.970 to 0.948 ± 0.016 and the EEG reference from 0.966 to 0.967 ± 0.019 , so neither pooled mean is an artifact of testing on animals the model has already seen. Two questions stay open. The first is scale: five folds of one cohort cannot settle transfer across laboratories, species or seizure-induction protocols, and multi-cohort subject-wise evaluation should become the default here as it already has in EEG decoding more broadly [28, 56]. The camera-only literature now spans mice [26, 45], further rodent preparations [21, 44] and even zebrafish [54], so cross-preparation transfer is testable with data that already exists. The second is fine-grained severity on unseen animals, the regime where video gives up the most (0.662 to 0.475 ± 0.024) and the one a deployed screen would actually face.

(2) Severity grading. Detection is largely solved; grading is where the headroom is. The error concentrates in the rare severe stages: Stage 4 and Stage 5 together account for only 201 of the 5,289 validation clips, and video reaches 0.36 and 0.42 per-class F1 on them (Table 17). Progress here depends less on architecture than on how those stages are represented and scored. Three directions follow. Targeted collection of severe events is the most direct and the most expensive. Ordinal-aware objectives are the cheapest: the flat softmax we use scores a Stage 2-for-Stage 5 confusion exactly as it scores Stage 2-for-Stage 3, and the conditional ordinal formulation of Sec. 3.2 drops that assumption without collecting anything new. Fusion is the third, recruiting the amplitude cues in the electrographic

trace that a camera cannot see. The imbalance strategy itself also deserves a controlled comparison: Sec. 5.3 reports inverse-frequency-weighted cross-entropy without contrasting it against unweighted training, resampling [11] or a focal objective [32], a well-mapped design space [22] that needs no new data; precision-recall summaries [46] would also suit the rare stages better than the threshold-free AUROC we report alongside macro-F1. One further caveat attaches to the target rather than the model: the Racine scale [43] was derived from electrically kindled seizures and its transfer to other induction models is not automatic [52], so some of the residual five-class error is plausibly annotation variance.

(3) Cross-modal artifact rejection. The clinical reason for recording both streams is that each vetoes the other’s artifacts: a grooming bout that mimics an ictal discharge on a single channel is obvious on camera, and an electrode transient during quiet rest is obvious in the trace. Computationally that rationale is almost unexploited. The handful of human systems that fuse both streams [1, 9, 31, 37] and the two animal attempts [35, 40] all target binary detection, and none is organized around artifact rejection as the objective. Adaptive, quality-aware fusion (weighting each stream by its momentary reliability [2, 25]) is the natural mechanism to realize it. We keep the two streams unfused here by design (Sec. 3.1), so we report this as future work. One property of our setup bounds what any such study could conclude from our numbers: the EEG reference is single-channel, and a richer montage would change both the artifact profile and the achievable ceiling.

(4) Benchmarks. RodEpi [42] provides the first public *video-only* rodent benchmark (13,053 clips, 19 subjects), and our backbone reaches 0.979 subject-wise F1 on it (App. E.2), which indicates the video result is not an artifact of our particular cohort. A shared *paired* video-EEG animal benchmark is still missing, and its absence is why the few fusion papers cannot be compared against one another or against their would-be animal successors. App. A.5 documents how severely the scarcity of paired, annotated data constrains the whole field. Extending a RodEpi-style release with synchronized EEG and stage-level labels is, in our view, the single most valuable contribution available here; we release a de-identified description of our cohort as a step toward it.

(5) Real-time and on-device inference. Preclinical monitoring runs continuously for weeks or months, so throughput decides whether a model is usable at all. vEpiNet’s 5.7 min of compute per recorded hour [31] shows that real-time video-EEG is already feasible in the human setting. Our backbone-agnostic result points the same way for animals: X3D-M matches R(2+1)D-18 while using 4M parameters against 33M (Table 2, Fig. 3), so a deployable model costs little accuracy. The streaming problem is what remains. Clip-level inference on pre-segmented events is a dif-

Row-normalised confusion matrices (fraction of each true class; counts in parentheses)

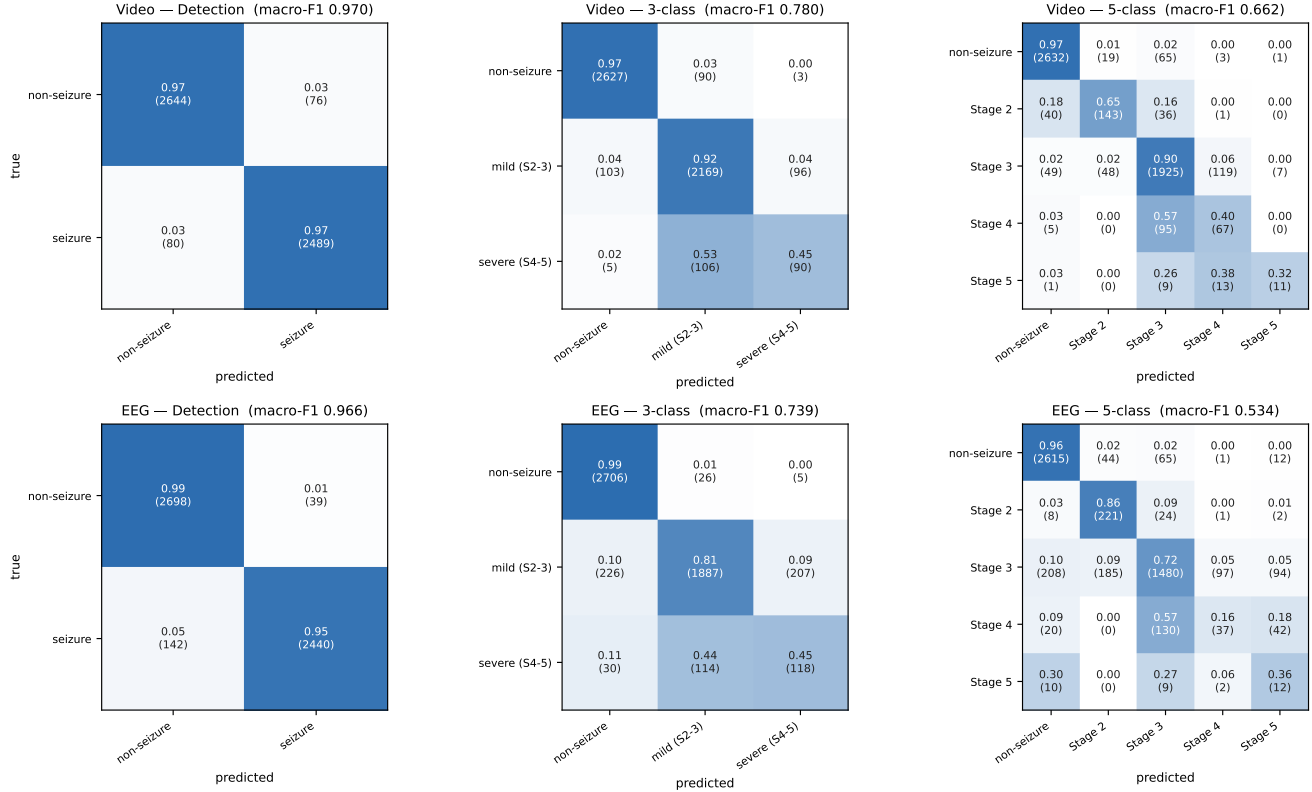


Figure 16. Row-normalised confusion matrices for both modalities at all three granularities (seed-42 runs; fraction of each true class, raw counts in parentheses). The 5-class panels carry the argument of App. E.4: video’s mass sits on the sub-diagonal next to the true class, whereas EEG’s leaks both to non-seizure and to non-adjacent stages.

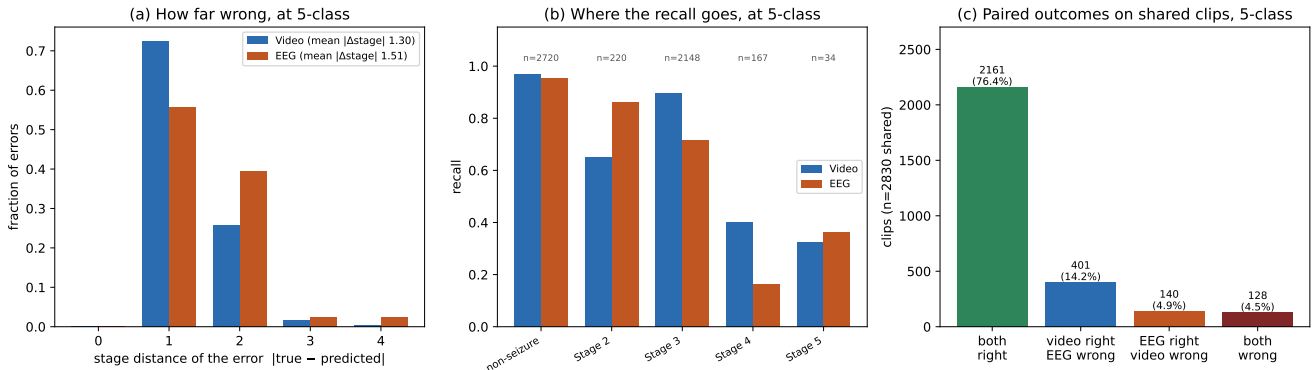


Figure 17. Error structure at 5-class. (a) Distribution of $|true - predicted|$ over errors: video’s are concentrated at one stage, EEG’s spread further (mean 1.30 vs. 1.51). (b) Per-class recall with support annotated; the ordering reverses at Stage 2. (c) Paired outcomes on the 2830 clips both models scored.

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ferent task from continuous detection over an unsegmented feed, where the operative metric becomes false positives per recorded hour, and where a detector must also decide event boundaries rather than receive them.